Searching for hidden attractors as a rare-event problem: An adaptive strategy portfolio with a worst-case guarantee

Maksim Elisov[a)]
Samara National Research University, Moskovskoye Shosse 34, Samara 443086, Russia

(Dated: 14 October 2026)

An attractor is hidden if its basin does not intersect small neighborhoods of the equilibria, so trajectories from unstable equilibria never reach it. When the basin is also small, locating it is a rare-event detection problem in the space of initial conditions. We characterize a search method by its detection probability $P(N)$ after N integrations and its cost $N_{0.9}$ relative to uniform sampling, which needs $2.3/p$ integrations for basin measure p; an adaptive method can gain only from what trajectories that miss the target reveal. We propose a portfolio of multilevel splitting driven by transient duration, active learning of basin boundaries, and a uniform share θ, which is at most $1/\theta$ times costlier than uniform sampling in integrations and inherits the scaling exponent of its splitting arm. On a four-level benchmark with a verification protocol, it reduces the cost of finding rare hidden attractors ($p \sim 10^{-3}$–10^{-2}) by factors of 3–7 where transients carry a signal; on a synthetic family, the cost scales as $p^{-0.45}$ down to $p = 10^{-5}$; on coupled copies of systems with hidden attractors, the gain reaches 10^3–10^4. In wall time, the gain persists for bounded trajectories (10–27 for a rare regime of two coupled Chua circuits) but vanishes where most trajectories escape quickly. A stagger-and-step chain on the same coordinate is as strong where a signal exists: the gain comes from the coordinate, and the portfolio supplies the bound. The protocol also reveals a coupling-born hidden chaotic attractor of the coupled circuits.

Some attractors of nonlinear systems are hidden: no trajectory started near an equilibrium ever reaches them, so the textbook way of finding attractors misses them. When such an attractor also has a tiny basin, finding it is like searching for a needle in a haystack of initial conditions, and published methods rarely report whether they beat random guessing. This paper recasts the search as a rare-event problem and measures the cost of finding a target with a prescribed confidence. This makes methods comparable and shows that an advantage can come only from what unsuccessful trajectories reveal about the basins. The resulting search strategy is guaranteed never to be much worse than random sampling in the number of trajectories, is several times better on typical problems, and is orders of magnitude better on rare targets in coupled systems. It comes with a benchmark and a verification protocol that catch numerical artifacts and transient chaos mistaken for attractors. Measured in computing time rather than in the number of trajectories, the advantage holds where trajectories stay bounded, as in dissipative physical systems.

I. INTRODUCTION

An attractor is called hidden if its basin of attraction does not intersect small neighborhoods of equilibria; otherwise, the attractor is self-excited.[1–3] Hidden oscillations occur in Chua circuits,[4–6] in control systems with saturation,[7] in the Aizerman and Kalman problems,[8] in generalized Lorenz and Rabinovich systems,[9–11] in systems without equilibria,[12,13] with a single stable equilibrium,[14,15] with lines and surfaces of equilibria,[16,17] in two-dimensional maps,[18–20] and in neural network models.[21] Reviews[22–24] show that the number of such examples has grown by orders of magnitude over the past fifteen years. Since a hidden attractor usually coexists with other attractors, searching for it is a problem of multistable dynamics.[25,26] Attractors whose basins occupy a small fraction of the state space are called rare.[27,28]

The main practical difficulty stems from the definition itself. The standard procedure for finding attractors launches trajectories from neighborhoods of unstable equilibria along their unstable manifolds. By definition, this procedure cannot find a hidden attractor, and an ad hoc choice of other initial conditions easily misses it. Analytical–numerical methods have been proposed for localizing hidden attractors: harmonic balance and the describing function followed by continuation in a parameter.[2,29,30] There are also geometric heuristics: perpetual points,[31–33] connecting curves[34–36] and surfaces,[37] coordinate-wise zero-velocity sets,[38] and parameter switching.[39] Evolutionary search over initial conditions is used as well.[40,41] A general tool is the estimation of basins by sampling initial conditions,[42–46] together with set-oriented methods.[47,48] A sample of n initial conditions detects an attractor of basin fraction p with probability $1-(1-p)^n$,[44] so rare attractors are found either by very large samples or by continuation from parameter values where they are not rare: path following combined with basin stability[49] and global continuation,

[a)]Electronic mail: maksimelisov2003@gmail.com

which seeds the search at each parameter value with the attractors found at the previous one.[44,50,51] The limitations of the heuristics are well known. Perpetual points may not exist at all, or they may lead to something other than the target.[52–54] Connecting curves do not always pass through the basin of the hidden attractor.[24]

Assessments of localization methods are almost always qualitative: the attractor was found from a certain point. How many integrations were needed, with what probability the method finds the target within a given budget, and whether it beats uniform sampling at all are rarely reported; the review in Ref. 24 and the evolutionary search of Ref. 40 are representative. Hiddenness and rarity are, moreover, different properties. In well-known examples, the basin of a hidden attractor often occupies most of the natural region of initial conditions (Sec. III). What makes the problem hard is a small basin measure p, not the absence of a connection to equilibria. In addition, a long-lived transient chaotic set is sometimes declared a "hidden attractor." Examples are the "hidden attractor" claimed for the classical Lorenz system[55,56] and for the Rabinovich–Fabrikant system;[57] see also Refs. 58–60. Comparing methods therefore requires an explicit verification protocol.

If the target basin is small, finding the attractor becomes the problem of detecting a rare event in the space of initial conditions. Well-developed tools exist for such problems. They include multilevel splitting[61–63] and its static counterpart, subset simulation,[64] the cross-entropy method,[65] and sequential design of experiments with active learning of the failure domain.[66,67] Novelty search and quality-diversity algorithms belong here as well.[68–71] Rare-event algorithms have been applied, for example, to extreme events and transitions in climate models[72,73] and to high-dimensional dynamical systems.[74] Machine-learning methods for basins of attraction[51,75–79] estimate the fractions of already known basins rather than search for unknown attractors. To the best of our knowledge, these tools have not been applied systematically to the search for hidden attractors.

The contributions of this work are as follows.

1. The search for a hidden attractor is formulated as a rare-event detection problem. The quality of a method is measured by the detection probability $P(N)$ as a function of the budget N, by the cost $N_{0.9}$, by the gain factor G relative to uniform sampling, and by the scaling exponent s in $N_{0.9} \propto p^{-s}$. We identify the three channels of information through which an adaptive method can outperform uniform sampling.

2. We build a four-level benchmark: three-dimensional flows, maps, four-dimensional systems, and synthetic families with analytically known basin measure, with search experiments on the first, second, and fourth levels. We develop a protocol for identifying and verifying attractors. Refining the protocol exposed an integration-step artifact in the Chua circuit and a discrepancy with a published table of basin fractions.

3. We propose a portfolio of adaptive strategies: multilevel splitting driven by transient duration, active learning of basin boundaries, and a uniform share. We show that in the number of integrations it is at most $1/\theta$ times costlier than uniform sampling and inherits the scaling exponent of the splitting arm. We construct a cost model with diagnostics that explain, given a known target, when a gain is possible.

4. We measure the gain and its scaling with basin measure and dimension, and the cost in integrator steps and wall time against uniform sampling with the same evaluation and with recurrence-based early stopping (Attractors.jl). The gain carries over to time where trajectories are bounded and shrinks where most of them escape quickly. The enumeration of regimes by the protocol reveals a hidden chaotic attractor of two coupled Chua circuits that the single circuit does not have. Failure cases are documented explicitly, including the limit set by the time resolution of the reaction coordinate.

II. PROBLEM FORMULATION

A. Hidden attractors

Consider an autonomous system in $\mathbb{R}^n$: a flow $\dot{x} = F(x)$ or a map $x_{k+1} = F(x_k)$ with a smooth (or piecewise-smooth) right-hand side. Let A be an attractor, $W(A)$ its basin, and $\mathcal{E}$ the set of equilibria (fixed points). The attractor A is *self-excited* if there exists $e \in \mathcal{E}$ such that $W(A) \cap U_\varepsilon(e) \neq \varnothing$ for every $\varepsilon > 0$; otherwise, A is *hidden*.[1] Three consequences matter for the search. Trajectories launched from neighborhoods of equilibria do not reach a hidden attractor. In systems without equilibria and in systems with a single stable equilibrium, all nontrivial attractors are hidden. A hidden attractor may coexist with self-excited ones, and then its basin is an "island" that the unstable manifolds of the equilibria do not reach.

A continuous characteristic complements the binary definition. Let ν_ε be the uniform distribution on the union of the ε-neighborhoods of the equilibria (in an extended version, also on the ε-tubes around their unstable manifolds). We call

$$h_\varepsilon(A) = \Pr_{x\sim\nu_\varepsilon}\left[x \in W(A)\right] \tag{1}$$

the *hiddenness measure*. For a hidden attractor and a finite set of equilibria, $h_\varepsilon(A) = 0$ for all sufficiently small ε, and the function $\varepsilon \mapsto h_\varepsilon$ shows the "depth" of hiddenness; for a continuum of equilibria, a common ε is not guaranteed, and for systems without equilibria we

set $h_\varepsilon = 0$. A measured $h_\varepsilon = 0$ is a necessary condition of hiddenness, not a sufficient one: it bounds the measure of $W(A) \cap U_\varepsilon(\mathcal{E})$ from above by the sampling resolution. Below, it is measured for the benchmark systems (Sec. III C).

B. Search as rare-event detection

Let $B \subset \mathbb{R}^n$ be a compact region of initial conditions with normalized Lebesgue measure μ, $\mu(B) = 1$. The target attractor A^* has a basin W^* of measure $p = \mu(W^* \cap B) \ll 1$. The unit of cost is one integration of standard length $T = T_{\mathrm{tr}} + T_{\mathrm{obs}}$ (a transient segment and an observation window); for maps, it is one orbit. Below, one integration of one initial condition is also called one *evaluation*. An algorithm $\mathcal{A}$ chooses initial points $x_1, x_2, \cdots \in B$ adaptively: the point x_k may depend on the results of the previous integrations. Let $\tau = \min\{k : x_k \in W^*\}$ be the index of the first hit. The quality of the algorithm is described by the detection probability and by the cost at confidence level $1 - \delta$:

$$\begin{aligned} &P_{\mathcal{A}}(N) = \Pr[\tau \le N], \\ &N_{1-\delta}(\mathcal{A}) = \min\{N : P_{\mathcal{A}}(N) \ge 1 - \delta\}. \end{aligned} \tag{2}$$

The probability is taken over the internal randomness of the algorithm; throughout, $\delta = 0.1$. The reference point is uniform independent sampling:

$$\begin{aligned} &P_0(N) = 1 - (1-p)^N, \\ &N^0_{1-\delta} = \left\lceil \frac{\ln \delta}{\ln(1-p)} \right\rceil \approx \frac{\ln(1/\delta)}{p}, \qquad N^0_{0.9} \approx \frac{2.3}{p}. \end{aligned} \tag{3}$$

The gain factor $G_{1-\delta}(\mathcal{A}) = N^0_{1-\delta}/N_{1-\delta}(\mathcal{A})$ is equivalent to an "effective basin measure" $p_{\mathrm{eff}} = pG$. On families with a controlled basin measure, we measure the scaling exponent s in $N_{0.9} \propto p^{-s}$; for uniform sampling, $s = 1$. A method has an *order-of-growth gain* if $s < 1$ with statistical significance and a *constant-factor gain* if $s \approx 1$ but $G > 1$. A hit counts only if the candidate passes the identification and verification protocol of Sec. III C; transient chaos mistaken for an attractor is a failure of the method.

C. Where a gain can come from

If an algorithm receives only the target indicator of each point, adaptivity cannot help in the worst case. Partition B into $\lfloor 1/p \rfloor$ cells of measure p and let W^* be a cell chosen uniformly at random. An algorithm that learns only whether $x_k \in W^*$ excludes at most one cell per query, so $\Pr[\tau \le N] \le Np$ and $N_{1-\delta} \ge (1-\delta)/p$ for every algorithm; uniform sampling is therefore within the factor $\ln(1/\delta)/(1-\delta) \approx 2.6$ of the optimum. A gain is possible only through the information carried by the trajectories that did *not* land in W^*. Among the sources of such information that cost one integration per point, we distinguish three channels (information about the vector field that needs no integration, such as harmonic balance or Lyapunov functions, is outside this setting).

- **Channel A: basin geometry.** Labels are spatially correlated. If the target basin adjoins already found boundaries between known basins, sampling near these boundaries is enriched with the target.
- **Channel B: transients.** Trajectories converging to known attractors or escaping to infinity carry information about proximity to unseen invariant sets: the escape time, delays near saddles, and the mismatch between the early segment of a trajectory and its asymptotic regime. Near a basin boundary, a trajectory follows for a long time the stable manifold of the invariant set on the boundary (the edge state[80–82]), and its transient lengthens. Invariant sets on basin boundaries are classically located by straddle (bisection) methods that exploit exactly this effect,[83] and long transients are the signal of the stagger-and-step method for finding chaotic saddles.[84]
- **Channel C: prior sets.** Classical heuristics (nullclines, perpetual points, connecting curves) define a distribution of starting points that may be enriched with the target.

The proposed method (Sec. IV) uses all three channels. The cost model (Sec. IV E) indicates which channel works for which basin properties and provides an observable diagnostic.

III. BENCHMARK AND IDENTIFICATION PROTOCOL

A. Levels of the benchmark

The benchmark is designed to separate two properties, hiddenness and rarity, and to test the method on all types of systems for which heuristics have been proposed. Equations, parameters, and integration settings are given in Appendix A, and a summary is given in Table I. A gallery of level-A attractors is shown in Fig. 1.

- **Level A: three-dimensional flows.** The Chua circuit with two parameter sets[2,6,24] (Fig. 2 and Appendix A). Systems without equilibria: NE4, Wang–Chen #1, and Wei,[12,13] as well as the 2026 system of Jafari *et al.* without fixed and perpetual points.[54] Systems with a single stable equilibrium: Wang–Chen #2, SE3, and SE11.[14,15] The LE1 system with a line of equilibria[16] and the Rabinovich system.[10] The negative controls are the transient chaos of the Lorenz system at $(a, r, b) = (4, 29, 2)$[58] and the "hidden attractor" of the Rabinovich system at $r = 6.8$, which, according to Ref. 56, is a transient set.

- **Level B: maps.** The map of Huynh *et al.* without fixed points.[20] Eight cases of the general family of Jiang *et al.*:[18] four without fixed points and four with a single stable one. Three parameter sets of the same family in the form of Ref. 85. The Hénon map serves as a control. Phase–basin portraits of the maps and of the other systems are given in Appendix A.
- **Level C: four-dimensional systems.** The simplified Lorenz system of Li and Sprott without equilibria[86] in three regimes (a torus and a pair of strange attractors; a cycle and a pair; hyperchaos) and the system of Jahanshahi *et al.*[87] This level is used for the census and the verification protocol; the search experiments are run on levels A, B, and D.
- **Level D: synthetic families** with analytically known basin measure (Sec. III B).

The region B for three-dimensional systems is the cube $[-10,10]^3$, the natural region of the original publications. For the Chua circuit, we take the box $[-10,10]^2\times[-15,15]$ proposed in Ref. 40. For NE4 and the Rabinovich system, the region is twice the extent of the attractor. Basin fractions are estimated from 5×10^4 uniform points for level A and 2×10^4 for the other levels; for the Chua circuit, they were checked by an independent computation in Attractors.jl.[43,88]

Table I confirms that hiddenness and rarity are different properties. In the Rabinovich system and in Wang–Chen #2, the hidden attractor occupies 76% and 67% of the region. Only the targets of Chua II, SE11, LE1, the Jafari cycle and chaos, and the Huynh map are rare ($p<10^{-2}$). The Jiang maps, which Ref. 18 characterizes as having "tiny" basins, give $p\approx0.03$–0.10 in boxes around the attractor. The classical heuristics behave on this benchmark as described in the review in Ref. 24, and none of them helps on the rare targets (Appendix C). The literature baseline for the search experiments is differential evolution (DE) over initial conditions with the cost function of Ref. 40, which, unlike our strategies, is given the size of the target (Appendix D).

B. Synthetic families with known basin measure

To measure the scaling of the cost for p down to 10^{-7} and dimensions up to 14, the three-dimensional system SE3 with a hidden attractor ($p_W=0.153$ in $[-10,10]^3$, where W_{HA} denotes its basin) is embedded in $\mathbb{R}^{3+m}$. The search region is $B=[-10,10]^3\times[-1,1]^m$. We use two constructions with opposite properties: D1 has no transient signal, and D2 has a signal on all scales.

D1, "embedding with a gate." The system is

$$\dot{x}=F(x)-g(w)\,(x-e),\qquad \dot{w}=-\kappa w,$$
$$g(w)=\frac{H}{1+\exp\bigl(-400(|w|-\rho)/\rho\bigr)}, \tag{4}$$

where e is the stable equilibrium of the base, $H=50$, and $\kappa=0.5$. While $|w|>\rho$, the "gate" pulls x toward e. Therefore, the basin of the hidden attractor is approximately $W_{\mathrm{HA}}\times\{|w_0|\le\rho\}$, and

$$p(\rho,m)=p_W\,\frac{V_m(\rho)}{2^m},\qquad V_m(\rho)=\frac{\pi^{m/2}\rho^m}{\Gamma(m/2+1)}. \tag{5}$$

The measured fractions agree with Eq. (5) within 5%–15%. Outside a thin rim of the ball, the transient hardly depends on $|w_0|$, so channel B is practically absent in D1.

D2, "edge state." Here, the embedding is decoupled:

$$\dot{x}=F(x),\qquad \dot{w}=\kappa\,w\,\bigl(|w|^2/\rho^2-1\bigr),\qquad \kappa=1. \tag{6}$$

The sphere $|w|=\rho$ is repelling. For $|w_0|<\rho$, the coordinates w decay and x follows the base; for $|w_0|>\rho$, the norm $|w|$ blows up in finite time. The target basin is exactly $W_{\mathrm{HA}}\times\{|w_0|<\rho\}$, and formula (5) is exact given p_W; a check gave $p_{\mathrm{meas}}=0.00102$ and 0.00100 at $p=0.001$. The escape time for $|w_0|=\rho(1+\Delta)$ equals $\ln\bigl((1+\Delta)^2/((1+\Delta)^2-1)\bigr)/(2\kappa)\approx\ln(1/2\Delta)/(2\kappa)$ for small Δ. It diverges at the basin boundary (Fig. 3), as for edge states of real systems.[82,89] In D2, channel B is informative on all scales down to the integration step.

C. Identification and verification protocol

Trajectory fingerprint. A trajectory is integrated by the fourth-order Runge–Kutta method (RK4) with a fixed step (maps are iterated) over the transient segment T_{tr} and the observation window T_{obs}. Escape is detected by the condition $|x|>R_{\mathrm{esc}}$. On the observation window, we compute the largest Lyapunov exponent λ by the two-trajectory method,[90] the mean speed, the maximum norm, the frequency of local maxima, the amplitude contraction ratio, and the coordinate-wise mean, standard deviation, minimum, and maximum. The same statistics are computed on an early window; they are needed by the reaction coordinate (Sec. IV A). The kernels are implemented in Numba[91] and compute the fingerprints inside the integrator, at about 3×10^8 RK4 steps per second on 14 cores.

Templates and matching. The template of a regime is the mean μ and spread σ of the fingerprint over 64 copies. The copies start from random points of a trajectory segment on the attractor of length $4T_{\mathrm{obs}}$ (phase randomization) with relative noise 10^{-3}. A trajectory with fingerprint f is assigned to the template with the smallest

$$z=\max_i\frac{|f_i-\mu_i|}{\sigma_i+0.02\,(1+|\mu_i|)}, \tag{7}$$

provided that $z<6$ and the classes are compatible: a chaotic template requires $\lambda>0.005$, and a regular one requires $\lambda<0.02$; in the overlap, the fingerprint decides. Converged trajectories are compared with equilibria by

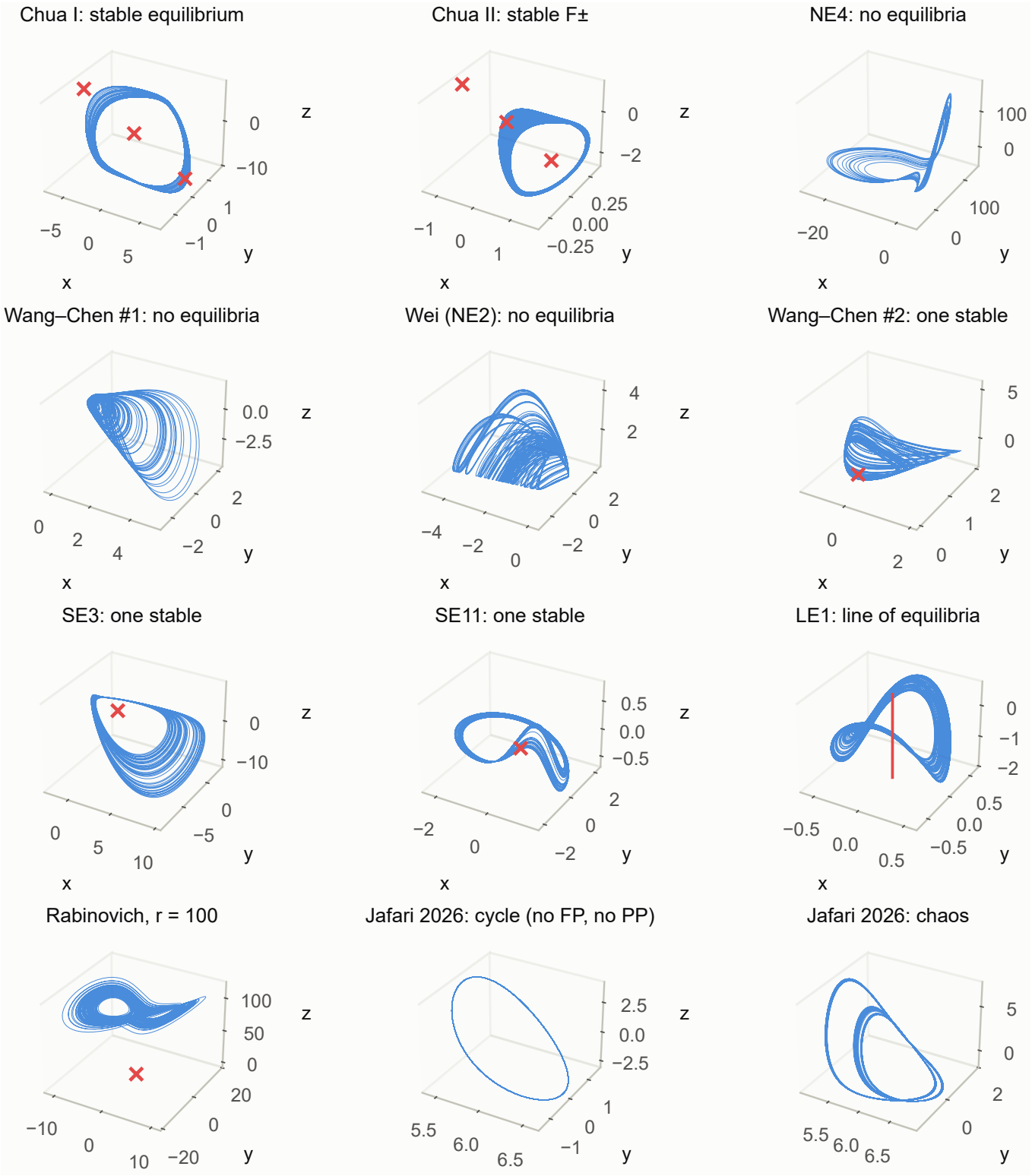


FIG. 1. Hidden attractors of the level-A benchmark. Crosses mark equilibria; for LE1, the red line is the line of equilibria $x = y = 0$. NE4, Wang–Chen #1, Wei, and the Jafari 2026 system have no equilibria, and in Wang–Chen #2, SE3, and SE11 the single equilibrium is stable, so all attractors shown are hidden by definition.

TABLE I. Benchmark: search targets (hidden attractors), basin measures p in the region B, and the cost of uniform sampling $N^0_{0.9}$. Basin measures are from the uniform census (5×10^4 points for level A, 2×10^4 otherwise); $N^0_{0.9}$ is the 0.9 quantile of the first-hit index over $R=50$ runs of uniform sampling ($R=20$ for the Jafari chaos; for the nonrare targets, it is $\lceil \ln 0.1/\ln(1-p)\rceil$). A slash separates two symmetric attractors, a semicolon separates different targets, and "both" means the cost of hitting either of the two; the search tables of Sec. V use the combined measure of the two symmetric attractors and remeasure $N^0_{0.9}$ with their own runs. λ is the largest Lyapunov exponent.

Level	System	Equilibria	Target	p	$N^0_{0.9}$
A	Chua I	F_0 stable, $F_\pm$ saddle-foci	two chaotic (A, B)	0.017 / 0.018	136 / 124
	Chua II	F_0 saddle-focus, $F_\pm$ stable	cycle LC; two chaotic	0.992; 0.0014 / 0.0013	1; 772 (both)
	NE4	none	chaos ($\lambda\approx0.03$)	0.33	6
	Wang–Chen #1	none	chaos	0.103	22
	Wei (NE2)	none	chaos	0.096	23
	Jafari 2026, $a_3=-1$	none; no perpetual points	cycle	0.009	229
	Jafari 2026, $a_3=-3.548$	none	fragile chaos ($\lambda\approx0.05$)	(2–5) $\times10^{-4}$	5331
	Wang–Chen #2	one, stable	chaos	0.666	3
	SE3	one, stable	chaos	0.153	14
	SE11	one, stable	chaos	0.0079	302
	LE1	line $x=y=0$	chaos	0.0016	1422
	Rabinovich, $r=100$	saddle at 0, two stable foci	chaos	0.763	2
B	Huynh 2019	none	chaos	0.0091	211
	Jiang 2016, NF I e, f, g, i	none	chaos	0.028–0.039	59–82
	Jiang 2016, SF I a–d	one, stable	chaos	0.065–0.10	22–37
C	Li–Sprott 4D, $a=6$	none	torus; two strange	0.090; 0.446 / 0.464	25; 4
D	D1, D2 (base SE3), $n=4$–14	as for the base	chaos of the base	10^{-2}–10^{-7}	258–2.1×10^7

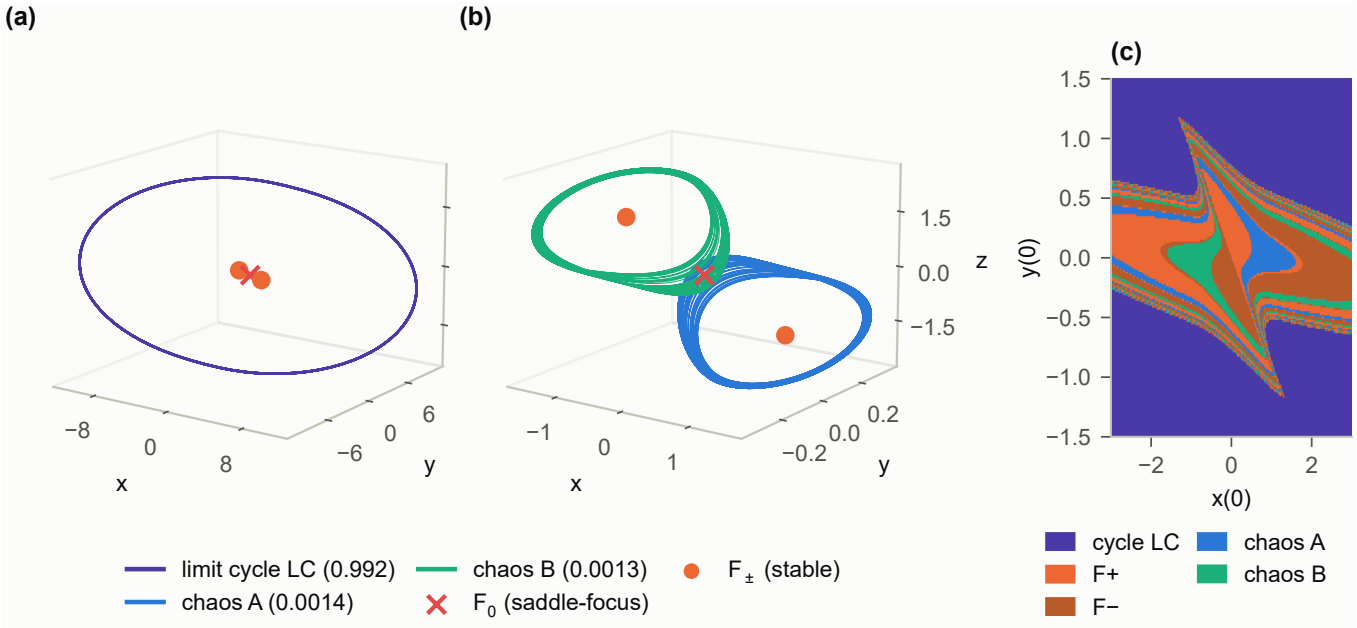


FIG. 2. Chua circuit, parameter set II: five coexisting attractors. (a) The large limit cycle LC (basin fraction in the box 0.992), the chaotic attractors A and B (0.0014 and 0.0013), the saddle-focus F_0, and the stable foci $F_\pm$. (b) Enlarged fragment; the chaotic attractors surround $F_\pm$. (c) Basins in the plane $z(0)=0$ (window $|x|\le3$, $|y|\le1.5$, axes not to scale). The basins of the chaotic attractors are narrow strips between the basins of $F_\pm$ and of the cycle; a search can exploit only these boundaries and the long transients near them.

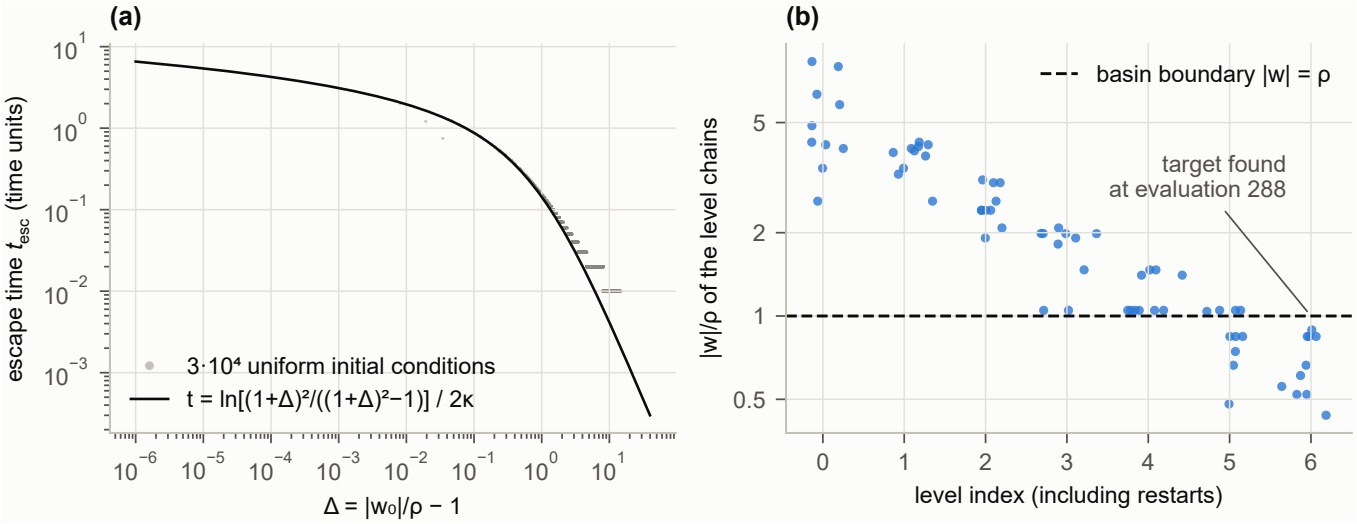


FIG. 3. Mechanism of the splitting gain in family D2 ($n = 6$, $p = 10^{-4}$). (a) Escape time versus the relative distance to the basin boundary $\Delta = |w_0|/\rho - 1$; dots are 3×10^4 uniform initial conditions, and the line is the exact law $\ln\big((1+\Delta)^2/((1+\Delta)^2-1)\big)/(2\kappa)$. The steps on the right reflect the discreteness of the integration step. (b) $|w|/\rho$ for the seeds of the chains over the levels of one run; the chains contract to the ball $|w| < \rho$, and the target is found after 288 integrations, whereas uniform sampling needs $N^0_{0.9} = 1.7 \times 10^4$.

their final point with tolerance $10^{-3}L$, where L is half the largest edge of B. Decaying trajectories ($\lambda < -0.002$ and amplitude contraction) are compared by the center of the window. Unmatched trajectories are clustered (DBSCAN), and a new template is calibrated for each cluster; this is how unknown regimes are discovered, during a search as well as in the enumeration of the regimes of a new system.

Verification. Each candidate passes four checks: the lifetime (the integration is continued for at least 20 windows and, for claims of a chaotic attractor, up to 10^4–10^5 time units), convergence in the step (the same template and stable basin fractions under $\Delta t \to \Delta t/2$), an independent integrator (DOP853[92] with rtol $= 10^{-11}$ and Attractors.jl with Vern9), and the sign of the Lyapunov exponent over consecutive windows. For claims of hiddenness, h_ε is estimated by sampling in the ε-neighborhoods of all equilibria and along their unstable manifolds, $\varepsilon \in \{10^{-3}, 10^{-2}, 10^{-1}\}L$; it distinguishes deeply hidden attractors (SE3, SE11) from shallowly hidden ones (Chua I, the Rabinovich system), and the number of starts bounds its resolution (Appendix B). The protocol was refined on three cases that change the conclusions if missed (Appendix B): a numerical "transient chaos" of the Chua circuit at $\Delta t = 0.01$, which underestimates the fractions of the hidden attractors almost twofold; a published table of basin fractions whose hidden-attractor fractions coincide with ours only at that coarse step; and intermittency and long-lived transient sets mistaken for attractors. The step-convergence and lifetime checks are therefore mandatory.

IV. METHOD

The method is a portfolio of three arms: multilevel splitting driven by transient duration (channel B), active learning of basin boundaries (channel A), and a uniform share that bounds the worst case. Prior sets (channel C) enter as a distribution of starting points and candidates. All arms receive, for each point, only its fingerprint and the label produced by the protocol of Sec. III C. The strategies do not know which label is the target. Two configurations are used below: the *portfolio with full integration*, with which the gains in integrations of Sec. V B–V D are measured, and the *portfolio with two-stage evaluation* of Sec. IV F, with which the costs in steps and wall time of Sec. V F are measured.

A. Reaction coordinate: transient duration

For an initial point x_0 with a trajectory fingerprint, we define

$$\xi(x_0) = \begin{cases} \dfrac{\ln k_{\mathrm{esc}}}{\ln N_{\mathrm{tot}}} \in [0,1], & \text{escape at step } k_{\mathrm{esc}}, \\ 1 + D(x_0), & \text{bounded trajectory}, \end{cases} \tag{8}$$

where

$$D = \frac{\big\|(m_{\mathrm{early}} - m_{\mathrm{obs}},\ s_{\mathrm{early}} - s_{\mathrm{obs}})\big\|}{L}.$$

Here, $k_{\mathrm{esc}} \in \{1, \ldots, N_{\mathrm{tot}}\}$ is the integration step at which the trajectory left the escape radius, N_{tot} is the number of integration steps, and m and s are the coordinate-wise means and standard deviations over the early window (the first steps of the transient segment) and over

the observation window; all trajectories that escape at the first step share the value $\xi = 0$. A trajectory that quickly settles onto its regime has $D \approx 0$. A trajectory that lingers near an edge state on a basin boundary has a large D or a large k_{esc}. In rare-event algorithms, the levels are defined by a score (importance) function, and the optimal score is the committor, the probability of reaching the target before the alternatives.[73,93] The committor is unknown here; ξ is a cheap proxy computed from a single trajectory that does not use the target label. The coordinate is built on transient rather than asymptotic features because only the former carry a signal (Appendix G): on uniform samples, transient features separate the points near the target basin from the rest with an area under the receiver operating characteristic curve (AUC) of 0.88–0.97, whereas asymptotic features (the Lyapunov exponent, the range) reach only 0.5–0.75.

B. Multilevel splitting in the space of initial conditions

For the static problem "initial condition → fingerprint," the static counterpart of adaptive multilevel splitting[62] is subset simulation;[64] below, we call it splitting. Levels are set adaptively by a quantile of ξ, and the sample above a level is replenished by conditional Markov chains. The algorithm is as follows.

1. Reference sample: $n_0 = 50$ uniform points of B.
2. The level ξ_l is the value of ξ at the $(n_0 p_0)$th largest point, $p_0 = 0.2$. These $n_0 p_0 = 10$ points are the seeds of the chains.
3. Each chain makes $1/p_0 - 1 = 4$ random-walk steps $y = x + \sigma(b-a)\eta$, $\eta \sim \mathcal{N}(0, I)$, with mirror reflection into the parallelepiped $B = [a, b]$. A step is accepted if $\xi(y) \geq \xi_l$; a rejected proposal leaves the chain in its current state, which is counted again as a state of the level. Every proposal costs one integration, so a level costs $n_0(1-p_0) = 40$ integrations. For the uniform measure on B, a symmetric proposal with reflection, and a fixed σ, this is the Metropolis algorithm for the conditional distribution on $\{\xi \geq \xi_l\}$.
4. The step σ (initial value 0.1) is adapted to the acceptance rate, a simple variant of adaptive proposal scaling in subset simulation:[94] below an acceptance rate of 0.2, it is halved; below 0.4, it is divided by 1.25; above 0.6, it is multiplied by 1.5.
5. The next level is chosen from all states of the chains of the current level (step 2). On stagnation (a relative increase of the level below 2% or $\sigma < 0.005$) for two levels, and also after 10 levels, the algorithm restarts with a new uniform reference sample.

Restarts protect against the case in which the chains have contracted onto a nontarget region with a long transient.

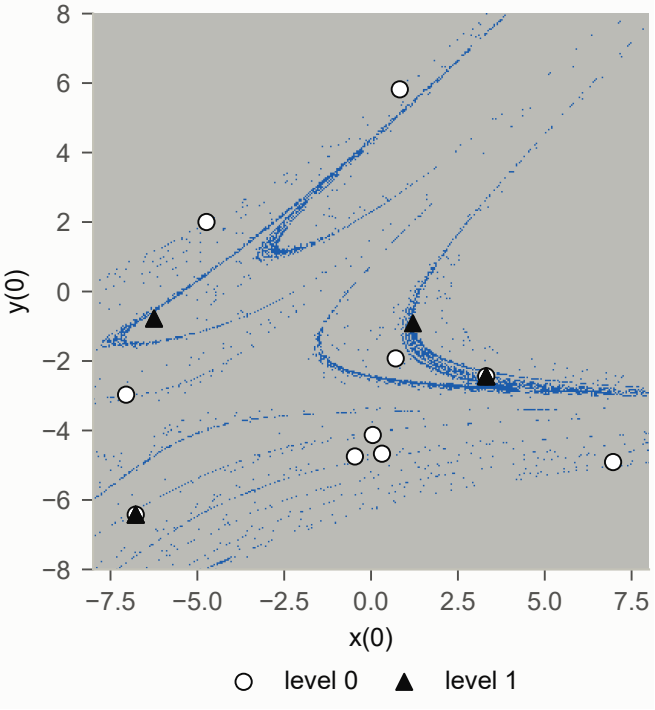


FIG. 4. Splitting chains on the basin map of the Jiang map NF I e (gray: escape; blue: basin of the chaotic attractor): seeds of levels 0 (open circles) and 1 (triangles). Already at the second level, the chains sit on thin "tongues" of the basin, since the escape step is larger near their edges; the target is found after 90 integrations. Seeds that coincide are drawn once.

In the benchmark, we compared three coordinates: ξ from Eq. (8), the escape step alone, and the multidimensional "rarity" of the fingerprint (the mean distance to the $k = 5$ nearest neighbors in the reference sample). Figure 4 shows how the chains of two levels are distributed over the basin map of a Jiang map, and Fig. 3 shows how the chains in D2 contract to the ball $|w| < \rho$.

C. Active learning of basin boundaries

The active-learning arm stores all evaluated points with their labels; escaping trajectories form a separate class. On these data, a k-nearest-neighbor classifier ($k = 10$, weights inversely proportional to distance) is trained in normalized coordinates of B. From a pool of 2000 candidates, the points with the largest entropy of the predicted class probabilities are selected for evaluation (uncertainty sampling[95]). Half of the pool consists of Gaussian perturbations of "boundary" points of the sample, i.e., points that have a point of another class among their k neighbors, with a scale of half the distance to the nearest neighbor of another class. The other half consists of uniform points or points of a prior set (nullclines). The classifier is retrained when the sample grows by 10%. Two implementation choices keep the cost of

this arm bounded (Appendix H): neighbors are found by brute-force search (for $n \geq 6$, or while the sample has at most 2000 points), which returns the same queries several times faster than a tree, and in the two-stage configuration the classifier is trained on the last 2×10^4 evaluated points only. While the data contain a single class, there are no boundaries, and the arm is idle.

D. Portfolio and worst-case guarantee

In each round, the portfolio evaluates a batch of points from the splitting arm (10 points), from the active-learning arm (20 points if the data contain at least two classes), and uniform points. The number of uniform points is $\lceil \theta/(1-\theta) \cdot n_{\mathrm{arm}} \rceil$, where n_{arm} is the number of points from the arms and $\theta = 0.2$ is the uniform share. Points within a batch are shuffled so that no arm is favored in the accounting of the first-hit time. The reference sample is shared by both arms. An arm is called *autonomous* if the choice of its points depends only on the results of its own evaluations. The uniform arm and the splitting arm are autonomous; the active-learning arm uses all data. The uniform share plays the role of the defensive component of mixture importance sampling,[96] and combining strategies with different failure modes follows the idea of algorithm portfolios, which reduce the risk of heavy-tailed run times.[97,98]

Proposition 1 *Let c be the maximum batch size, ω_j the guaranteed share of arm j in each round, and $N_{1-\delta}(j)$ the cost of arm j run on its own. Then*

$$N_{1-\delta}(\mathit{portfolio}) \leq \min\left\{ \frac{N^0_{1-\delta} + c}{\theta}, \min_{j \ \mathit{autonomous}} \frac{N_{1-\delta}(j) + c}{\omega_j} \right\}. \tag{9}$$

Proof. After N evaluations of the portfolio, an autonomous arm j has performed at least $\omega_j N - c$ evaluations of its own. By autonomy, the sequence of its points is distributed as in a stand-alone run with the same generator. Therefore, $\Pr[\tau > N] \leq \Pr[\tau_j > \omega_j N - c] \leq \delta$ whenever $\omega_j N - c \geq N_{1-\delta}(j)$. For the uniform arm, $N_{1-\delta}(j) = N^0_{1-\delta}$ and $\omega_j = \theta$. ∎

The proposition is an elementary property of a defensive mixture, and its value is in the construction it justifies. Its consequences are as follows. First, in the number of integrations, the portfolio is at most $1/\theta = 5$ times costlier than uniform sampling, up to the additive term c/θ: $G \geq \theta N^0/(N^0 + c)$, which for $c = 38$ and $N^0 \approx 200$ is 0.17 rather than 0.2. A single adaptive method has no such bound: splitting alone, and novelty search driven by asymptotic features (Appendix D), fail to find the target within the budget in some runs. Second, the scaling exponent of the portfolio is no larger than that of the best autonomous arm: a constant factor $1/\omega_j$ is lost, but not the order. In our implementation, the share of the splitting arm is $\omega \approx 0.26$ when the active-learning arm is active and $\omega \approx 0.77$ without it. Two qualifications apply. The bound holds in integrations, not in integrator steps or wall time: the arms select long transients, so a point of the portfolio may cost more steps than a uniform point (Sec. V F). Autonomy of the splitting arm is exact with full integration; with two-stage evaluation and templates that grow during the search (Sec. IV F), the coordinate of its points depends weakly on what the other arms have found, so the inheritance of its exponent is then approximate, and the bound through the uniform arm requires that stage 1 never stops a target trajectory, which is verified in Sec. V F.

E. Cost model and channel diagnostics

Splitting with selection fraction p_0 builds nested sets $S_l = \{\xi \geq \xi_l\}$ with $\mu(S_{l+1} \mid S_l) \approx p_0$; one level costs $n_0(1-p_0)$ evaluations. Denote by $\pi_l = \mu(W^* \mid S_l)$ the fraction of the target in the lth set and by $e_l = \pi_{l+1}/\pi_l \in [0, 1/p_0]$ the enrichment at level l; $e_l < 1$ means that the coordinate is deceptive at that level. The target becomes frequent when $\prod_{l<K} e_l \approx 1/p$, and the cost is $N \approx n_0 + n_0(1-p_0)K + O(1/\pi_K)$.

Proposition 2 (cost model) *Assume that the chains sample the conditional distribution on each S_l exactly, that the number of levels is unbounded, and that the final stage samples S_K independently. (i) If the coordinate is ideal, $e_l = 1/p_0$, then $K = \lceil \ln(1/p)/\ln(1/p_0) \rceil$ and $N = O(n_0 \ln(1/p))$, i.e., $s = 0$. (ii) If the enrichment is constant, $e_l = p_0^{-\zeta}$ with $0 < \zeta \leq 1$, then $K \approx \ln(1/p)/(\zeta \ln(1/p_0))$, and the cost is still logarithmic. (iii) If the coordinate is informative only down to a depth K^* ($e_l = 1$ for $l > K^*$), then $N \approx n_0 + n_0(1-p_0)K^* + 1/\pi_{K^*+1}$: the scaling exponent returns to $s = 1$, with a constant-factor gain $\prod_{l \leq K^*} e_l$. (iv) If the coordinate is deceptive, $e_l < 1$, the same count gives a constant factor below 1, i.e., a loss.*

Proposition 2 follows from counting levels. It is a model of the mechanism rather than a statement about the implementation: the implementation restarts after ten levels, so the logarithmic regime is available only while $p \geq p_0^{10\zeta}$; the states of a chain are correlated samples, which is why a coordinate with zero informative depth gives $G < 1$ rather than $G = 1$ (family D1, Sec. V C); and the exponent $s \approx 0.45$ observed on D2 over $p = 10^{-2}$–10^{-5} (Sec. V C) is a crossover between the logarithmic regime at $p \gtrsim 10^{-4}$ and the discretization limit below it, not an asymptotic law. Its content lies in the classification: an order-of-growth gain requires a coordinate that is informative on all scales, a finite informative depth yields a constant-factor gain, and a deceptive coordinate yields a loss. The transient duration diverges at a basin boundary, where the trajectory follows the stable manifold of the edge state. If the target

basin is surrounded by such a boundary and the sets S_l are nested all the way down to it, the enrichment persists on all scales (the case of D2). If the boundary of the target does not stand out against other boundaries, or if the duration saturates, only the near zone is informative. Saturation occurs, for example, for the discrete escape times of maps and for escape by slow drift.

Diagnostics from the uniform sample. The enrichment e_l is not observable before the target is detected, but on a benchmark system with a known target, it can be estimated from a uniform sample. The decisive quantity is not the fraction of target points in the upper set S_q (the fraction q of points with the largest ξ) but spatial convergence toward the target. Let $d(q)$ be the median distance (in units of L) from the nontarget points of S_q to the nearest target point of the sample. Splitting gains if $d(q)$ decreases as q decreases: the chains move by a local random walk and need a connected "path" of ever longer transients leading to the basin. If $d(q)$ does not decrease, the long trajectories are scattered over the region, and there is no gain even if the fraction of the target in S_q grows (Appendix G).

Discretization limit. The coordinate is built from a trajectory with step Δt, so escape times shorter than one step are indistinguishable. In D2, the escape time for $|w_0| \gg \rho$ equals $\rho^2/(2\kappa|w_0|^2)$, and the coordinate is informative only for $|w_0| \lesssim \rho/\sqrt{2\kappa\Delta t}$. The size of this zone relative to the region sets the limit of the gain at small p and small m (Sec. V C, Appendix F).

F. Two-stage evaluation

Most of the integration budget goes to trajectories that settle early onto a known regime yet are integrated over the full horizon, which the protocol requires only for target candidates (Sec. III C). We therefore evaluate every initial condition in two stages, for all strategies including uniform sampling. Stage 1 integrates the transient segment up to the end of the early window, $\min(n_{\rm tr}, n_{\rm obs})$ steps, where $n_{\rm tr}$ and $n_{\rm obs}$ are the numbers of steps of the transient segment and of the observation window, followed by a short observation window of $n_{\rm obs}/4$ steps. The trajectory stops after stage 1 if, on this window, it is identified with confidence as a known nontarget regime: an equilibrium (converged, or decaying with an amplitude below $0.02L$), or a known cycle or chaotic attractor with a z score below half the identification tolerance. Escaping trajectories are final anyway. All other trajectories (target candidates and unknown or ambiguous ones) continue from the end of the early window to the full horizon. Since the main trajectory and the early window are the same, their fingerprints coincide bit for bit with those of a single full integration, and the target is always confirmed by the full protocol. A stopped trajectory keeps its label; only its ξ is computed on the shorter window.

For continued trajectories, stage 1 costs extra steps, because a quarter of the observation window is integrated twice. A rule based on step counts therefore keeps the two-stage mode only while it pays. After the first 200 evaluations, the mode stays on as long as the steps saved over the last ten batches exceed the extra steps; otherwise, only every tenth batch is evaluated in two stages, as a probe. The rule uses step counts rather than time, so runs remain reproducible. On uniform samples from eight problems, the two-stage labels and the fingerprints of all continued and escaping trajectories coincided with those of full integration bit for bit, and no target hit was lost (Sec. V F).

V. RESULTS

A. Measurement protocol

Each system–strategy pair was run R times with fixed generator seeds: $R = 100$ for levels A and B, the synthetic families, and the ablation; $R = 50$ for the hyperparameter, horizon, bisection, and enlarged-region tests; and $R = 10$–100 for the coupled systems. A run stopped at the first hit of the target or when the budget was exhausted. Common random numbers were used: the first 50 points of all strategies coincide with the first 50 points of uniform sampling, so differences arise only after the initial sample. $N_{0.9}$ is the linearly interpolated 0.9 quantile of the first-hit index over the runs (infinity if fewer than 90% of the runs find the target). Confidence intervals are 95% bootstrap percentile intervals[99] (2000 resamples of the runs of both the strategy and the uniform reference). The exponent s is the slope of the regression of $\ln N_{0.9}$ on $\ln(1/p)$ over four values of p, with its interval from the same bootstrap.

Within each experiment, $N^0_{0.9}$ is remeasured by uniform sampling with the same R and seeds, which is why the uniform costs of Tables I, II, and III differ by sampling error; the gains of a table are always relative to the uniform runs of the same experiment. The cost tables of Sec. V F, whose runs are few ($R = 20$), use instead a fixed reference per system: $N^0_{0.9}$ of Table II ($R = 100$) or the exact $\ln 10/p$ where p is known exactly. We compare uniform sampling, Zelinka's DE (Appendix D), splitting by ξ from Eq. (8), active learning with the 20% uniform share, the portfolio, and the portfolio with nullclines (channel C), all with full integration. The two-stage configuration of the portfolio (Sec. IV F) is used only in Sec. V F; on runs shorter than 2×10^4 evaluations, its cap on the training sample changes nothing. Unless stated otherwise, a gain is the ratio of integration counts.

Uniform sampling for the coupled systems and the enlarged regions was run on a GPU with a kernel that reproduces the CPU trajectories and fingerprints bit for bit. The code was written with the assistance of the AI assistant Claude (Anthropic; used through Claude Code with the models Claude Fable 5.1 and Claude Opus 5.5, 2026) and verified by the bit-exactness tests between the

CPU and GPU kernels, by the independent integrators of Sec. III C, and by the agreement of its labels with Attractors.jl (Sec. V F).

B. Rare targets of levels A and B

On the three rare targets with a transient signal, Chua II, SE11, and LE1, the portfolio outperforms uniform sampling by factors of 7.4, 3.1, and 6.2, with the lower bound of the 95% interval above 1 (Table II, Fig. 5); with nullcline sampling, it also gains on Chua I (4.4 times). On LE1, the best single arm is splitting (8.3 times), whereas active learning loses to uniform sampling (0.48): the long "continuum of equilibria/escape" boundary attracts the queries, and the target adjoins it only along a small part. The portfolio limits this degradation. On three problems, the portfolio gains little or nothing: the Jafari cycle, the Jafari chaos, and the Huynh map. The cost model explains these failures (Sec. IV E; they are cataloged in Sec. V G), and the portfolio stays within the bound of Proposition 1 ($G \gtrsim \theta = 0.2$): 0.43, 0.42, and 1.40. Zelinka's DE, which is given the size of the target, is nowhere significantly better than uniform sampling; on the Jafari chaos and cycle, it loses by factors of 4.3 and 3.4. On the Jiang maps ($p \approx 0.03$–0.07), all strategies are close to uniform sampling: at such p, the target is often found within the initial sample, and adaptation has nothing to work with. A single-chain stagger-and-step climb on the same coordinate[84] is stronger than the splitting arm and than the portfolio where the signal is present (10 on Chua II) and, without a uniform share, falls below the bound of the portfolio where the coordinate is deceptive or absent (0.16 on the Jafari cycle; Appendix D): the gain comes from the coordinate, and the role of the portfolio is the bound. The contributions of the arms (ablation), the comparison with bisection toward basin boundaries and with stagger-and-step, and the robustness to hyperparameters and to the integration horizon are given in Appendix D; no hyperparameter variant brings the gain below 1, and scaling both integration windows by 0.5 or 2 keeps the gains within a factor of 1.7 of the base values.

C. Scaling with basin measure and dimension

On family D2 ($n = 6$), the gain grows as p decreases from 10^{-2} to 10^{-5} (Table III, Fig. 6). The scaling exponent of the portfolio is $s = 0.45$ [0.42, 0.51], and that of splitting by ξ is 0.41 [0.37, 0.46], against 0.99 [0.93, 1.04] for active learning and 0.95 [0.90, 1.03] for uniform sampling (theoretical value 1). On a problem with a signal on all scales, the method thus gains in order of growth: at $p = 10^{-5}$, the gain is 93 for splitting and 72 for the portfolio. The exponent is a crossover value over this range rather than the asymptotic $s = 0$ of the ideal cost model (Sec. IV E); a single stagger-and-step chain on the same coordinate reaches $s = 0.32$ [0.28, 0.37] (Appendix D). On family D1, where there is no signal outside the thin rim of the ball, all strategies have s from 0.67 to 0.98; splitting does not help ($G = 0.49$–0.70), and active learning gives a factor of 1.1–3.5 that grows as p decreases, because escapes inside the ball serve as boundary markers. This is the prediction of Proposition 2 for a coordinate with zero informative depth, with the loss below 1 coming from the correlation of the chain states.

Below $p = 10^{-5}$, the gain on D2 breaks down: at $p = 10^{-6}$, within 2×10^5 integrations, the portfolio finds the target in 35% of runs against 18% for uniform sampling, and at 10^{-7}, within 10^6 integrations, in 16% against 9.5%. The breakdown is the discretization limit of Sec. IV E: the coordinate is informative only in the zone $|w_0| \lesssim \rho/\sqrt{2\kappa\Delta t}$, which occupies 2.3% of the region at $p = 10^{-5}$, 0.23% at 10^{-6}, and 0.023% at 10^{-7}, so an initial sample of 50 points contains an informative point with probability 0.69, 0.11, and 0.011, and the level threshold of the chains then degenerates (Appendix F). Reducing the integration step restores the gain: fourfold at $p = 10^{-6}$ gives $G = 450$ for the portfolio and 530 for splitting, and sixteenfold at $p = 10^{-7}$ gives 1200 and 1400, with all runs successful. The limit is set by the time resolution of the reaction coordinate rather than by splitting. The same limit explains the dependence on dimension at fixed $p = 10^{-4}$ (Appendix F): the portfolio gains from 17 times at $n = 6$ to 45–61 times at $n = 10$–14, but nothing at $n = 4$, where the basin in the additional coordinate has width $\rho = 6.6 \times 10^{-4}$ and the coordinate is flat on 99.5% of the region.

D. Coupled systems: rare hidden targets

Coupled copies of SE11. Coupling several copies of a system with a hidden attractor yields a real system with a rare hidden target: the regime in which every copy sits on its hidden attractor. We take k copies of SE11 with mean-field coupling along y:

$$\dot{\boldsymbol{x}}_u = \boldsymbol{F}(\boldsymbol{x}_u) + \epsilon\,(\bar{y} - y_u)\,\boldsymbol{e}_y, \qquad u = 1, \dots, k, \tag{10}$$

where $\bar{y} = k^{-1}\sum_v y_v$ and $\epsilon = 0.01$, in the region B^k. The only equilibrium is the synchronous equilibrium of SE11, which is stable, so the regime "all copies on the hidden attractor" is hidden. Its basin measure is 5.5×10^{-5} for $k = 2$ ($n = 6$) and 3.5×10^{-7} for $k = 3$ ($n = 9$), close to p_1^k for independent copies ($p_1 = 0.0079$). Here the portfolio gains by orders of magnitude (Table IV): 65 [42, 87] at $n = 6$ and 1400 [900, 2400] at $n = 9$, where it needs about 5×10^3 integrations instead of 7.6×10^6; all runs found the target. The transient duration is set by the copies that have not yet settled, so successive levels of splitting fix the copies one after another, and the cost grows roughly geometrically in k with a ratio far smaller than $1/p_1$. Without coupling, the gains are the same (1600 at $n = 9$), and with four uncoupled copies ($n = 12$, $p = p_1^4 = 3.9 \times 10^{-9}$), the portfolio finds the target in $N_{0.9} = 4.7 \times 10^4$ integrations, a gain of 1.3×10^4 relative to the exact

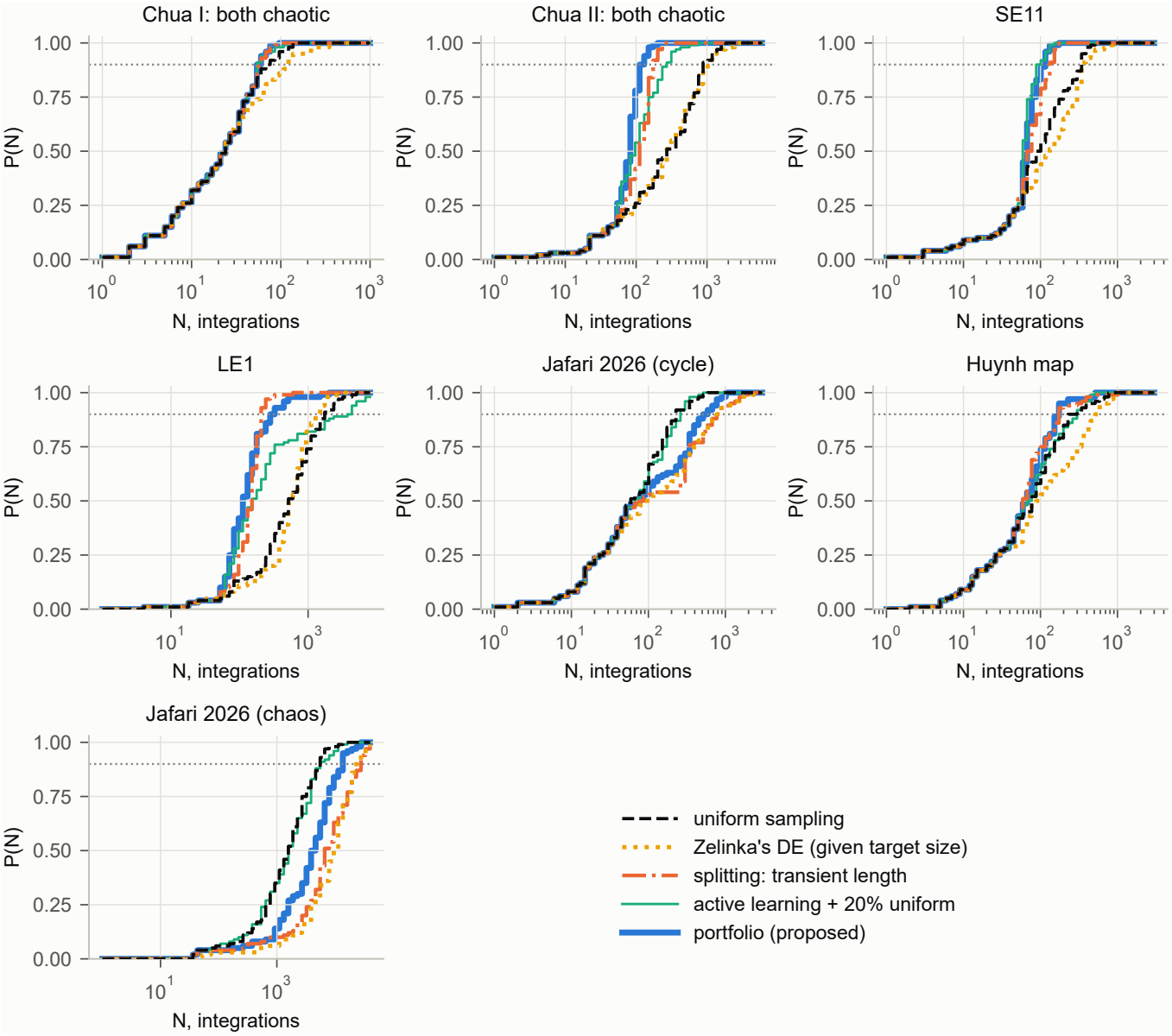


FIG. 5. Detection probability $P(N)$ as a function of the number of integrations for rare targets of levels A and B ($R = 100$ runs). The dotted line is the 0.9 level. For $N \leq 50$, all curves coincide by construction (common random numbers). On Chua II, SE11, and LE1, the portfolio curve (thick) lies to the left of uniform sampling over the whole range; on the Jafari cycle and chaos, it lies to the right, but within the guaranteed factor $1/\theta$.

TABLE II. Gain factors $G_{0.9} = N^0_{0.9}/N_{0.9}$ with 95% bootstrap intervals ($R = 100$); p is the combined measure of the two symmetric attractors where there are two. Bold: values whose lower interval bound exceeds 1. "—": the variant was not run. For the Jiang maps, the individual cases are given without intervals.

System	p	$N^0_{0.9}$	Zelinka's DE	Splitting	Active learning + 20% uniform	Portfolio	Portfolio + nullclines
Chua I	0.035	66	0.60 [0.43, 0.99]	1.20 [0.92, 1.77]	1.23 [0.95, 1.74]	1.24 [0.94, 1.77]	**4.42** [3.27, 6.47]
Chua II	0.0027	818	0.84 [0.67, 1.32]	**5.05** [4.13, 7.07]	**3.16** [2.58, 4.90]	**7.36** [6.09, 10.2]	**7.64** [5.98, 11.8]
SE11	0.0079	317	0.95 [0.72, 1.07]	**2.53** [1.95, 2.86]	**3.62** [2.63, 4.16]	**3.10** [2.46, 3.70]	**3.56** [2.71, 4.31]
LE1	0.0016	1617	1.31 [0.94, 2.08]	**8.29** [6.35, 12.3]	0.48 [0.32, 1.49]	**6.24** [3.76, 10.0]	**4.62** [3.03, 10.7]
Jafari 2026, cycle	0.009	219	0.29 [0.19, 0.50]	0.29 [0.19, 0.53]	0.88 [0.66, 1.41]	0.43 [0.29, 0.78]	0.32 [0.25, 0.50]
Huynh 2019	0.0091	221	0.45 [0.34, 0.85]	1.36 [0.79, 2.42]	0.82 [0.57, 1.72]	**1.40** [1.16, 2.54]	—
Jafari 2026, chaos	$(2–5) \times 10^{-4}$	5265	0.23 [0.16, 0.28]	0.22 [0.15, 0.28]	1.08 [0.57, 1.40]	0.42 [0.29, 0.58]	0.27 [0.16, 0.32]
Jiang NF I e/f/g	0.028–0.039	109/86/71	0.83/0.90/1.00	1.52/1.23/1.08	1.02/1.51/1.15	1.46/1.49/1.23	—
Jiang SF I a/b	0.065	36/33	0.75/0.92	1.00/1.00	1.00/1.00	1.00/1.00	—

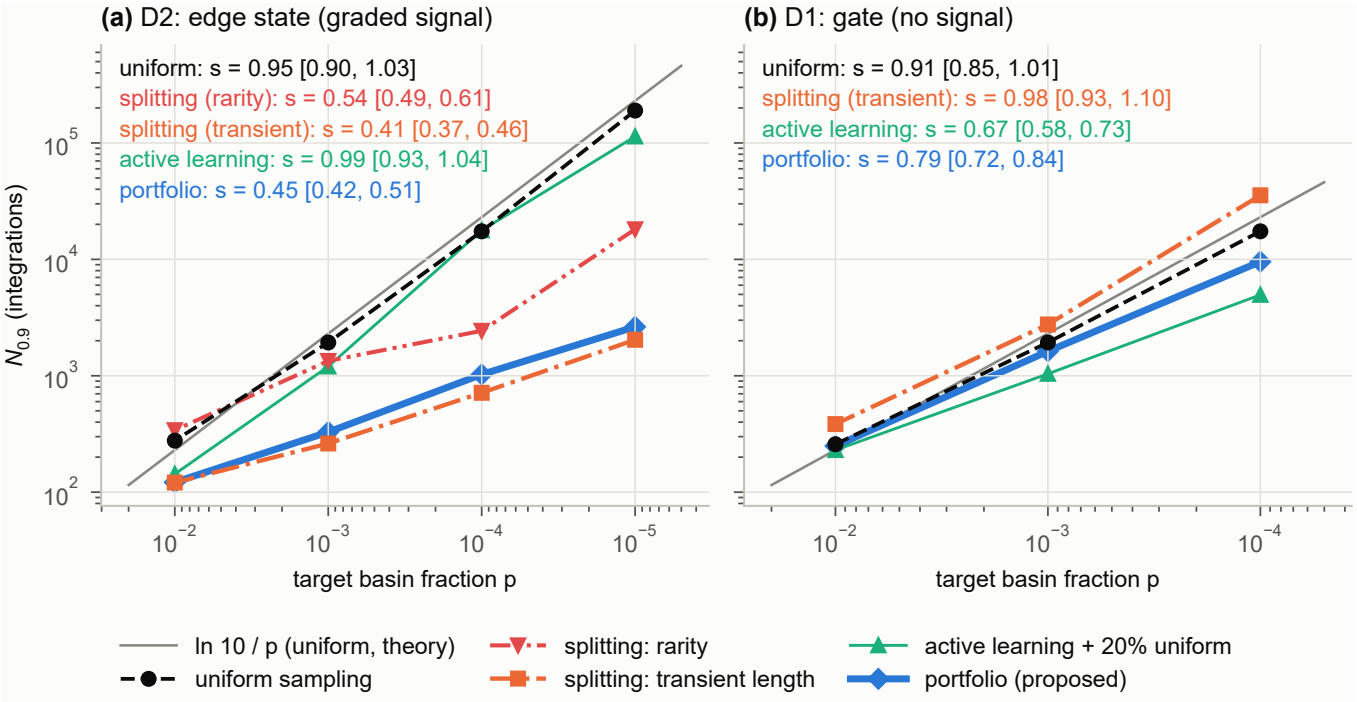


FIG. 6. Detection cost $N_{0.9}$ versus basin measure p ($n = 6$, $R = 100$); the legend gives the scaling exponents s, defined by $N_{0.9} \propto p^{-s}$, with 95% intervals. (a) Family D2 with a signal on all scales; splitting by ξ and the portfolio have $s \approx 0.4$–0.45 (splitting by the escape step alone coincides with splitting by ξ and is not shown; "rarity" is splitting by the multidimensional rarity of the fingerprint, Appendix D). (b) Family D1 without a signal; all strategies have $s \approx 0.7$–1. Thin gray line: $\ln 10/p$.

TABLE III. Family D2 ($n = 6$): cost of uniform sampling $N^0_{0.9}$ (in parentheses, the exact value $\ln 10/p$) and the costs and gains of splitting and of the portfolio ($R = 100$). For $p \le 10^{-6}$, the budget of the strategies (2×10^5 at $p = 10^{-6}$ and 10^6 at $p = 10^{-7}$) is smaller than $N^0_{0.9}$, and the fraction of runs that found the target within it is given, with that of uniform sampling in brackets; the last two rows give the gains with the integration step reduced 4 and 16 times (Appendix F).

p	Uniform	Splitting	Portfolio
10^{-2}	277 (230)	121 (2.3×)	122 (2.3×)
10^{-3}	1932 (2303)	262 (7.4×)	329 (5.9×)
10^{-4}	17 416 (23 026)	714 (24×)	1026 (17×)
10^{-5}	189 765 (230 259)	2039 (93×)	2634 (72×)
10^{-6}	2.19×10^6 (2.30×10^6)	24% [18%]	35% [18%]
10^{-7}	2.14×10^7 (2.30×10^7)	9% [9.5%]	16% [9.5%]
10^{-6}, $\Delta t/4$	(2.30×10^6)	530×	450×
10^{-7}, $\Delta t/16$	(2.30×10^7)	1400×	1200×

uniform cost $\ln 10/p$. Bisection toward basin boundaries does not scale on these systems: within budgets of $8/p$ at $n = 6$ and 2×10^5 at $n = 9$, it finds the target in 66% and 32% of runs. The coupling is weak, and the problem is close to a product of independent ones; if the attractors of a single copy are known, product initial conditions reach such a target directly, so the coupled copies test the mechanism of the gain rather than replace that shortcut.

Coupled Chua circuits. Most trajectories of SE11 escape, so its uniform points are cheap (Sec. V F); two mutually coupled Chua II circuits, with coupling $\epsilon(\bar{x}-x_u)$ in the first equation and $\epsilon = 0.1$, provide a system in which all trajectories are bounded. The nine equilibria (products of the equilibria of the single circuit) are found by Newton's method; the regimes of the coupled system are enumerated by the identification protocol on a uniform sample of 4×10^4 points, with the products of the attractors and stable equilibria of the single circuit as initial templates. The synchronized cycle LC takes 98% of the region. The rare hidden family consists of the regimes in which one circuit is on a hidden chaotic attractor and the other is at a stable equilibrium (and, at weaker coupling, both on hidden attractors); its total basin measure is 2.3×10^{-5}. None of the 2×10^5 trajectories per radius started in the $10^{-3}L$ and $10^{-2}L$ neighborhoods of the nine equilibria and along their unstable directions reaches it, so $h_\varepsilon < 1.5\times10^{-5}$ at the 95% level, below the measure of the family (zero hits would occur with probability 0.01 if h_ε were equal to p). On this target, the portfolio needs $N_{0.9} = 1900$ integrations against 1.05×10^5 for uniform sampling ($R = 20$, all runs successful), a gain of 55 in integrations; the gain in time is given in Sec. V F. Three uncoupled copies of Chua II give the extreme case. The regime "every copy on one of its two hidden attractors" has $p = (p_a+p_b)^3 = 1.9\times10^{-8}$ exactly, where p_a and p_b are the measures of the two attractors of the single circuit; this is beyond the reach of uniform sampling ($\ln 10/p = 1.2\times10^8$ integrations, about two weeks on four threads). The portfolio finds it in 9 of 10 runs within 4×10^5 integrations, with a median of 8.6×10^4 and $N_{0.9} = 3.2\times10^5$ (the largest successful run), a gain of 370 in integrations.

E. A hidden attractor born by the coupling

The same enumeration revealed a regime that the single circuit does not have (Fig. 7): one circuit stays on the large cycle LC, and the other hops chaotically between the regions of the two hidden attractors HA_a and HA_b, which in the single circuit at these parameters never merge.[5] Dynamically, it is the chaotic response of the second circuit to the near-periodic drive of the first, with the weak feedback of the coupling; chaos in a periodically driven Chua circuit is classical,[100] but here the drive is generated by the second copy of the same circuit, and the coupled system is autonomous. The regime passes the verification protocol. Sixteen copies (relative perturbation 10^{-3}) stay on it for 10^5 time units and four of them for 10^6 with RK4; eight copies integrated with DOP853 (rtol $= 10^{-11}$) for 2×10^4 time units end on the same template; it has the same template at $\Delta t/2$; and the largest Lyapunov exponent over windows of 100 time units is 0.095 ± 0.052. In 3%–6% of the windows, the fingerprint leaves the template during long stays of the second circuit near one of the former attractors, but no copy leaves the regime or decays to another attractor. It is hidden: none of the 2×10^5 trajectories per radius started in the $10^{-3}L$ and $10^{-2}L$ neighborhoods of the nine equilibria and along their unstable directions reaches it ($h_\varepsilon < 1.5\times10^{-5}$; at $10^{-1}L$, $h_\varepsilon = 0.18$, as for Chua I). Continuation in the coupling shows how it arises: for $\epsilon < 0.045$, the second circuit oscillates periodically near one of its stable equilibria (four symmetric periodic regimes); at $\epsilon \approx 0.045$, these branches merge into the chaotic hopping; λ grows almost linearly to 0.35 at $\epsilon = 0.46$, with periodic windows at $\epsilon = 0.09$ and 0.47; and at $\epsilon \approx 0.5$, the two branches merge with the synchronized cycle. Coupled Chua circuits with hidden attractors have been studied for synchronization,[5,6] and this regime does not appear to have been reported. It is hidden but not rare: its basin fraction is 0.4–1% per symmetric branch at all ϵ. Table V compares how fast the search methods find it: the standard procedure never does, uniform sampling and the portfolio find it within a few hundred integrations, and if the attractors of the single circuit are known, the product initial condition "LC × HA_a" lands on it in one integration, because the regime continues that product state. The value of the method here is the enumeration with verification rather than speed.

F. Cost in integrator steps and wall time

The number of integrations is a fair measure of cost only if all integrations cost about the same. They do not

TABLE IV. Coupled copies of SE11, Eq. (10): basin measure of the regime "all copies on the hidden attractor," cost of uniform sampling, and gains $G_{0.9}$ with 95% bootstrap intervals ($R = 100$ for $n = 3$ and for $n = 9$ with coupling; $R = 50$ otherwise). For $n = 12$, $N^0_{0.9} = \ln 10/p_1^4$ is the exact geometric value. Bold: values whose lower interval bound exceeds 1.

n	ϵ	p	$N^0_{0.9}$	Splitting	Portfolio
3	—	0.0079	317	**2.53** [1.95, 2.86]	**3.10** [2.46, 3.70]
6	0.01	5.5×10^{-5}	3.8×10^4	**44.0** [27.0, 65.6]	**64.9** [41.7, 86.6]
9	0.01	3.5×10^{-7}	7.6×10^6	**682** [448, 989]	**1435** [896, 2372]
9	0	4.9×10^{-7}	5.3×10^6	**598** [325, 1000]	**1574** [771, 2299]
12	0	3.9×10^{-9}	5.9×10^8	**4881** [3860, 6354]	**12 681** [9827, 18 098]

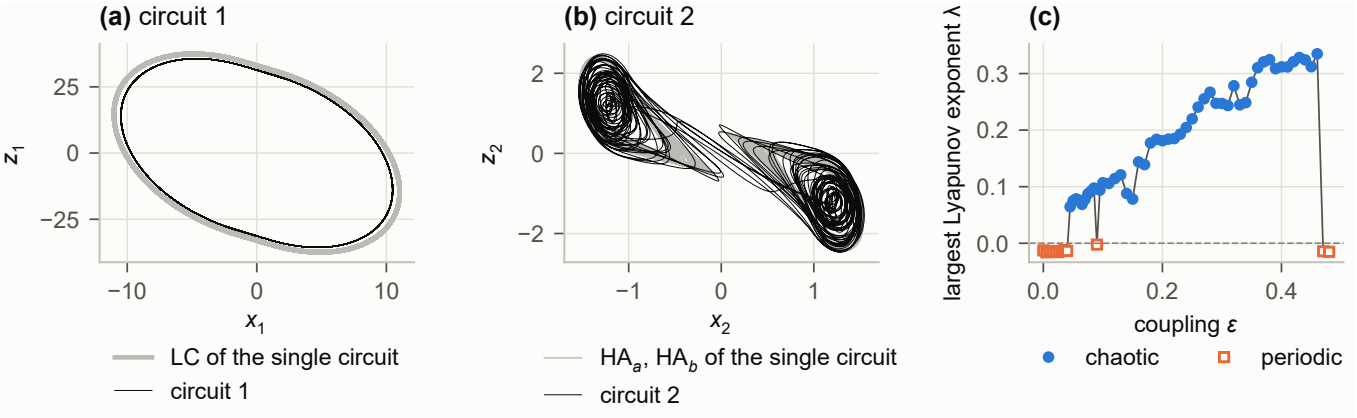


FIG. 7. Hidden attractor of two coupled Chua II circuits born by the coupling ($\epsilon = 0.1$). (a) Circuit 1 in the projection (x_1, z_1) against the cycle LC of the single circuit. (b) Circuit 2 in (x_2, z_2) against the hidden attractors HA_a and HA_b of the single circuit (gray): the trajectory hops between their regions. (c) Largest Lyapunov exponent of the regime versus the coupling, from continuation in ϵ: chaos is born at $\epsilon \approx 0.045$ by merging of periodic branches and disappears at $\epsilon \approx 0.5$.

TABLE V. Finding the coupling-born attractor of Fig. 7 (basin fraction 0.016 for the two symmetric branches): integrations to detection (0.9 quantile and median over $R = 20$ runs; for uniform sampling, the geometric value $\ln 10/p$ is 144) and wall time $T_{0.9}$ on one thread with full integration, two-stage evaluation, and recurrence-based stopping ("recur.," Attractors.jl, Sec. V F). "From equilibria": the standard procedure, which starts from the neighborhoods of the equilibria. The time of uniform sampling with two-stage evaluation and that of the portfolio are measured; the other two are rescaled from it by the measured ratio of costs per evaluation.

Method	$N_{0.9}$	Median	$T_{0.9}$, s		
			full	2-stage	recur.
From equilibria	never	—	—	—	—
Uniform	230	18	28	17	6.4
Portfolio	81	18	—	8.1	—
Product state[a]	1	1	0.07	—	—

[a]The initial condition "LC × HA_a" built from the attractors of the single circuit.

when trajectories escape: an escaping trajectory stops at the step of escape, and the method deliberately concentrates on long-lived trajectories. We therefore measured the cost of the two-stage configuration in integrator steps and wall time (Table VI, Fig. 8). The configuration is the three-arm portfolio with two-stage evaluation (Sec. IV F), brute-force neighbor search, and the capped training sample; for coupled systems, the products of the attractors of the single units other than the target are declared known ("product templates"). The reference is uniform sampling with the same two-stage evaluation at the fixed reference cost $N^0_{0.9}$ of Sec. V A, multiplied by the measured mean cost of a uniform evaluation. Steps are counted with the observation window twice, because the companion trajectory for the Lyapunov exponent is integrated alongside; steps and time up to the first hit are counted in proportion to the position of the hit within its batch, after an untimed warm-up that compiles the kernels. The gains are G_N in integrations, G_S in steps, and G_T in time ($R = 20$; three copies of Chua II: $R = 10$), with bootstrap intervals over the runs.

The problems fall into two groups. In the systems where most trajectories escape quickly (Chua I, SE11, LE1, D2, coupled SE11), the portfolio spends 2–300 times more steps per evaluation than uniform sampling, because it selects exactly the long transients that lead toward the target, whereas most uniform points escape within a few steps. The gain in integrations then shrinks in steps by a factor of 3–300: the portfolio is slower than uniform sampling in time on LE1 and on D2 at $p = 10^{-4}$ (upper interval bounds 0.76 and 0.54) and not faster on

TABLE VI. Cost of the two-stage configuration of the portfolio relative to uniform sampling with the same evaluation at the fixed reference cost of Sec. V A ($R = 20$; three copies of Chua II: $R = 10$, 9 successful runs): mean integrator steps per evaluation and gains in integrations (G_N), steps (G_S), and wall time (G_T, four threads per process) with 95% bootstrap intervals. Bold: values whose lower interval bound exceeds 1.

Problem	n	p	Steps per evaluation		G_N	G_S	G_T
			Uniform	Portfolio			
Chua I	3	0.033	2.6×10^4	4.6×10^4	**1.2** [1.1, 1.9]	0.36 [0.27, 1.4]	0.31 [0.24, 1.1]
SE11	3	0.0077	1139	3681	**4.5** [2.6, 5.3]	1.0 [0.84, 1.3]	0.85 [0.78, 1.1]
LE1	3	0.0013	334	1.1×10^4	**3.9** [1.2, 10]	0.095 [0.018, 0.65]	0.14 [0.029, 0.76]
D2, $p = 10^{-4}$	6	1.0×10^{-4}	14	1169	**17** [9.3, 33]	0.18 [0.12, 0.26]	0.26 [0.18, 0.54]
D2, $p = 10^{-5}$	6	1.0×10^{-5}	2	616	**100** [56, 210]	0.35 [0.052, 2.1]	1.5 [0.27, 3.1]
2 coupled SE11	6	5.5×10^{-5}	117	3694	**61** [36, 110]	**1.2** [1.1, 3.8]	1.1 [0.70, 3.0]
3 coupled SE11	9	3.5×10^{-7}	45	3418	**3600** [2400, 5200]	**41** [31, 57]	**29** [22, 44]
Chua II	3	0.0026	8.7×10^4	9.7×10^4	**7.3** [5.2, 9.6]	**6.1** [4.8, 8.6]	**4.1** [2.9, 6.3]
2 Chua II, $\epsilon = 0.1$	6	2.3×10^{-5}	3.0×10^5	3.4×10^5	**55** [24, 100]	**49** [20, 87]	**68** [14, 110]
2 Chua II, $\epsilon = 0$	6	7.3×10^{-6}	1.7×10^5	1.9×10^5	**190** [130, 290]	**160** [110, 230]	**130** [91, 200]
3 Chua II, $\epsilon = 0$	9	2.0×10^{-8}	1.7×10^5	2.3×10^5	360 [0, 1000]	270 [0, 740]	150 [0, 340]

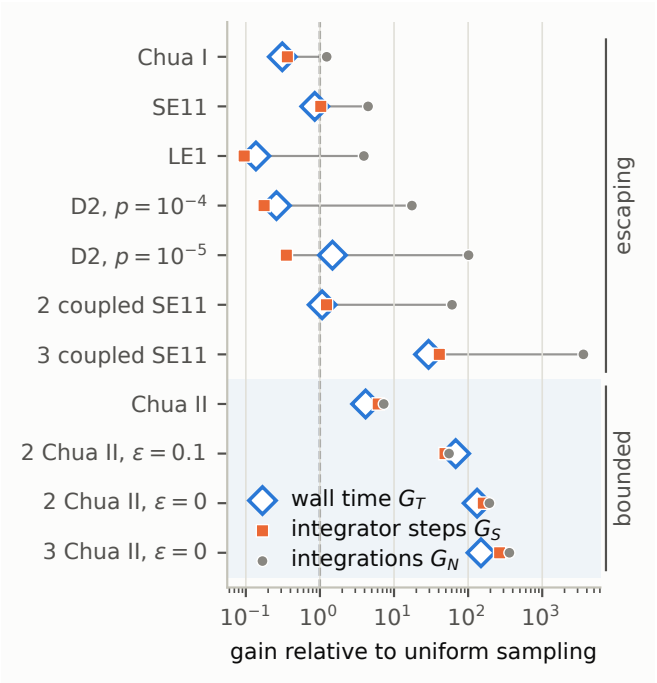


FIG. 8. Gain of the portfolio relative to uniform sampling in integrations (G_N), integrator steps (G_S), and wall time (G_T), both with two-stage evaluation, for the escaping systems (top group) and the bounded systems (bottom group). Dashed line: $G = 1$.

Chua I and SE11. It gains in time with statistical significance only where the gain in integrations is very large, on three coupled copies of SE11 ($G_N = 3600$, $G_T = 29$); on D2 at $p = 10^{-5}$ ($G_N = 100$), the time gain of 1.5 is not significant (interval [0.27, 3.1]). In the systems with bounded trajectories (Chua II and its coupled and uncoupled copies), every trajectory runs over the full horizon unless the two-stage evaluation stops it, the steps per evaluation of the portfolio and of uniform sampling are close, and the gain carries over to time: 4.1 on Chua II, 68 on the coupled circuits, and 130–150 on the uncoupled copies, all with lower interval bounds above 1 except for three uncoupled copies, where one run of ten failed. Two-stage evaluation makes both methods faster, uniform sampling more so: on Chua II, 94% of the uniform points and 91% of the points of the portfolio stop at stage 1; on uniform samples of 1500–4000 points from eight problems, it reduced the number of integrator steps by factors of 5.5 (Chua II), 2.2 (SE11), 1.9 (LE1), and 1.4 (two coupled copies of SE11), and on Chua I, the Jafari systems, and D2 the rule switches it off; no target hit was lost, and on Chua II the first-hit distribution of the portfolio agrees with that of full integration (mean over $R = 50$ runs: 74 against 70 integrations). The product templates matter for coupled systems: they let stage 1 stop the trajectories that settle onto product states other than the target and give the active-learning arm informative classes instead of "unknown." On three coupled copies of SE11, they lower $N_{0.9}$ from 3500 to 2100 and raise G_T from 9 to 29; they presuppose that the attractors of the single units are known, and the gain without them, 2200 in integrations and 9 in time, is the one that applies when they are not. The overhead of the arms themselves and the dependence of the time gains on the number of threads are given in Appendix H: the gains of Table VI were measured with four threads per process, and the one-thread values of Table VII are smaller by a factor of about 2.5 on the coupled circuits.

Comparison with recurrence-based evaluation. The cost of one evaluation can be reduced further by stopping a trajectory as soon as it recurs in the cells of an already known attractor, as in Attractors.jl.[43,44] For uniform sampling, the index of the first point in the target basin does not depend on how the trajectories are evaluated, so the time of uniform sampling with any evaluation scheme is $N_{0.9}^0$ times the mean cost of

one evaluation, and the schemes can be compared on a common set of initial conditions. Table VII compares, on one thread, full integration, our two-stage evaluation, and AttractorsViaRecurrences (Attractors.jl 2.0, sparse grid of 100 cells per dimension, sampling interval 0.1, adaptive Tsit5 with tolerance 10^{-9} for the Chua systems and fixed-step RK4 with our step for SE11 and LE1, whichever was faster), whose labels on the first 500 points coincide with ours and which recognizes the target states fed to it. On the bounded Chua systems with long transients, recurrence-based stopping is about 3 times cheaper per evaluation than our two-stage evaluation, which still needs the early window and a quarter of the observation window for the fingerprint; on the escaping systems, our evaluation is 2–8 times cheaper, because escape is detected by the radius at once. Against uniform sampling with recurrence-based stopping, the time gain of the portfolio on the coupled Chua circuits falls from 27 to 10, and on Chua II from 5.7 to 1.8, both still significant; the gain in integrations is untouched. The search strategy and the evaluation back-end are independent choices: the portfolio needs only a regime label and a transient duration for each point, both of which the recurrence mapper provides (Sec. VI).

G. Failures and limits

The failures below delineate the domain of applicability of the method.

1. **Deceptive coordinate.** In the Jafari 2026 chaos, almost all trajectories escape to infinity by slow drift, and the longest escapes lie in the region of slow drift rather than near the attractor; splitting finds the target 4.5 times more slowly than uniform sampling ($G = 0.22$), the case $e_l < 1$ of Proposition 2, and the portfolio stays within the bound (0.42). In an enlarged region, slowly returning trajectories make the coordinate deceptive even for Chua II: at $k = 4$, splitting alone loses by a factor of 7 and the portfolio gives 0.71 (Appendix E).

2. **Escape without a gradient of duration.** For the Jafari cycle, the escape time is determined by slow drift, and for the Huynh map, the escape times are discrete; the target is an "island" among escaping trajectories that the transients do not point to. No arm gains on the Jafari cycle, and the portfolio gains only 1.4 times on the Huynh map.

3. **A long boundary of known basins that the target touches only along a small part (LE1).** The active-learning arm has a median 3.7 times better than uniform sampling but a heavy tail: 14 of 100 runs need more than the uniform cost $N^0_{0.9} = 1617$, and two do not find the target within the budget. In the portfolio, this tail is bounded (median 109, maximum 1859 evaluations).

4. **Time resolution.** The discretization limit at very small p (D2, $p \leq 10^{-6}$; Sec. V C) and the cheap nontarget trajectories of escaping systems, where a gain of 2–120 in integrations is a draw or a loss in time (Sec. V F), are two faces of the same constraint: the coordinate and the cost are both set by the integrator steps, not by the integrations.

5. **Single adaptive strategies.** Splitting alone and the earlier strategies of Appendix D produce heavy tails or fail in a fraction of runs; the uniform share of the portfolio removes exactly these tails.

VI. DISCUSSION

What the rare-event formulation provides. The formulation in terms of $P(N)$ and $N_{0.9}$ turns the statement "the method finds a hidden attractor" into a measurable and comparable one. It makes visible three things that the qualitative setting obscures. First, many classical examples are not hard search problems, because the basin of the hidden attractor occupies a noticeable part of the natural region (Table I). Second, on truly rare targets the heuristics, including evolutionary search with Zelinka's cost function,[40] are no better than uniform sampling and sometimes worse (Table II). Uniform sampling is itself a strong reference: its cost $2.3/p$ is known in advance, does not depend on the geometry of the basin, and is within a factor of 2.6 of the best possible label-only algorithm (Sec. II C). Third, rarity is a property of the chosen region rather than of the attractor: enlarging the region or moving the parameter toward the crisis of the attractor lowers the basin measure by one or two orders of magnitude at most (Appendix E). A truly rare hidden target in a real system arises from a combination of several hidden regimes, as in coupled units (Sec. V D).

When adaptive search gains. A gain is possible only through the information carried by trajectories that missed the target (Sec. II C), and the data confirm this. Channel B (transients) gives an order-of-growth gain if the transient duration grows on approach to the basin on all scales, as in D2 and, judging by the diagnostic $d(q)$, in basins surrounded by edge states (Chua II, LE1, SE11); the enlarged regions of Appendix E give effective exponents of 0.4–0.7 on SE11, LE1, and Chua I, with the caveat that the geometry of the basin changes with the region. The most favorable case is a system of weakly coupled units with hidden attractors: the transient duration reports which units have not yet settled, splitting fixes them one after another, and the gain grows geometrically with the number of units. The order-of-growth gain is bounded in p by the time resolution of the coordinate, so the time step is a parameter of the search, not only of the integration. Channel A (boundaries) works if the boundary of the known basins is short and the target forms a noticeable part of it. Channel C (prior sets) gives a gain when the heuristic is fortunate (nullclines for

TABLE VII. Uniform sampling with three evaluation schemes against the portfolio, on one thread ($R = 20$): cost of one evaluation (ms), cost $N_{0.9}$ and time to detection $T_{0.9}$ (s) of the portfolio, and its time gain relative to uniform sampling with two-stage evaluation (G_T) and with recurrence-based stopping (G_T^{rec}), with 95% bootstrap intervals. For uniform sampling, $T_{0.9}$ is the fixed reference cost $N_{0.9}^0$ of Sec. V A times the cost of one evaluation; for the portfolio, it is the measured 0.9 quantile of the run time. Attractors.jl: the fastest of the tested settings. Bold: values whose lower interval bound exceeds 1.

Problem	p	$N_{0.9}^0$	Cost per evaluation, ms			Portfolio		G_T	G_T^{rec}
			full	two-stage	Attractors.jl	$N_{0.9}$	$T_{0.9}$, s		
Chua II	0.0026	818	26.7	6.07	1.95	113	0.87	**5.7** [4.7, 9.1]	**1.8** [1.5, 2.9]
2 Chua II, $\epsilon = 0.1$	2.3×10^{-5}	1.05×10^5	72.8	44.8	16.4	1899	172	**27.3** [11.4, 47.9]	**10.0** [4.2, 17.6]
SE11	0.0077	317	0.131	0.0562	0.131	71	0.028	0.64 [0.4, 0.71]	1.5 [0.93, 1.6]
LE1	0.0013	1617	0.021	0.0108	0.0819	413	0.58	0.03 [0.0057, 0.17]	0.23 [0.043, 1.3]

Chua I) and costs nothing when it is not. The diagnostic $d(q)$ separates the cases of gain and failure on all seven problems where it was measured, but it requires a known target and therefore serves as a tool of analysis rather than prediction. The splitting arm is a multichain relative of the stagger-and-step method,[84] which climbs the escape time of single trajectories to locate chaotic saddles. A single-chain climb on the same coordinate is as efficient as splitting or more where the signal is present, including the coupled copies and D2 down to $p = 10^{-5}$, and, without a uniform share, falls below the bound of the portfolio where the coordinate is deceptive or uninformative (Appendix D). What matters for the gain is the coordinate, not the particular way of climbing it, and a stagger-and-step chain is a natural arm of the portfolio, under the same guarantee.

When the gain is a gain in time. Counting integrations treats a trajectory that escapes after a few steps and one that runs over the whole horizon as equal. The method selects the long ones by construction, so the count overstates its advantage wherever most trajectories escape quickly, and there the gain in integrations is a loss or a draw in time (Sec. V F). Where trajectories are bounded, as in dissipative physical systems, a nontarget integration is as expensive as a target one, and the gain carries over to steps and to time. In time, the gain also depends on how fast the reference evaluates a trajectory: against uniform sampling with recurrence-based stopping, which needs no observation window, it is 2–3 times smaller than against uniform sampling with our two-stage evaluation. The two choices are independent: the portfolio needs from each trajectory only a regime label and a transient duration, which the recurrence mapper provides, so the next step is to combine them. The method thus pays in time when nontarget integrations are not cheap compared with the long transients it selects, or when the target is so rare that the gain in integrations exceeds the ratio of their costs.

Why a portfolio is needed. In the worst cases, single adaptive strategies lose to uniform sampling several times and produce heavy tails of the hit-time distribution (Sec. V G). By Proposition 1, the portfolio with a uniform share θ is guaranteed to cost no more than $1/\theta$ times uniform sampling in integrations and inherits the order of the best autonomous arm. On all failure problems, the portfolio stayed within these limits, whereas splitting alone fell well below them (Sec. V G, Appendix E), and the uniform share costs little on the favorable problems (Appendix D). Bisection toward basin boundaries[80,83] is a strong baseline on systems with smooth boundaries and costs nothing where there is no signal, but it loses where the signal is graded over many scales and does not scale to coupled units (Appendix D); as a fourth arm, it helps only where it is strong on its own.

Relation to continuation and to compositional search. The established way to find a rare attractor is to continue it from parameter values where it is easy to find, by path following or by global continuation.[44,49,50] This requires such a path; our method works at fixed parameters, which is the situation when a new system is explored or when the attractor exists only in a narrow parameter window. The two approaches are complementary: attractors found at neighboring parameter values are natural prior sets for channel C, and the portfolio can replace the uniform sample within each continuation step. For coupled units whose individual attractors are known, product initial conditions are another prior set, and in the coupled Chua circuits they reach both the rare family and the coupling-born attractor directly; the portfolio is for the case when the units' attractors are not known or the regime is not a product.

Verification as part of the method. Three episodes of this work (Appendix B) show that without a verification protocol a comparison of search methods loses its meaning. False "hidden attractors" due to transient chaos are known;[56,57] false "absences" due to the integration step are, according to our data, just as real. The same protocol, applied to enumerate the regimes of a new system, found the coupling-born attractor of the Chua circuits (Sec. V E). The hiddenness measure h_ε complements the binary definition; what matters in practice is the scale at which neighborhoods of the equilibria cease to lead to the attractor, and for rare targets the number of starts must be large enough for a zero to be informative (Sec. V D).

Limitations. All conclusions about gains refer to the benchmark, the synthetic families, and the coupled copies. On other systems, a gain is not guaranteed; only

the worst-case bound of Proposition 1 is, and only in integrations. For maps and for systems with escape by slow drift, the reaction coordinate may be uninformative or deceptive, and in a large region, slowly returning trajectories make it deceptive even for Chua II. The discretization limit bounds the dynamic range of the coordinate by the time step. The gains of 10^3–10^4 in integrations are obtained on weakly coupled or uncoupled copies, whose rare target is close to a product of the units' attractors; in that setting, declaring the product states of the units known improves the gain, and the numbers without this prior knowledge are reported alongside (Sec. V F). The same formulation applies to trained models of dynamics, such as reservoir computers in closed loop,[101,102] which may develop attractors absent from the training data;[103–106] that application will be reported separately.

VII. CONCLUSIONS

Formulating the search for hidden attractors as rare-event detection makes search methods comparable: the cost $N_{0.9}$ relative to uniform sampling is a single number that any method can report, and uniform sampling itself is a reference whose cost $2.3/p$ is known in advance and is within a constant of the best label-only algorithm. The benchmark, the identification and verification protocol, and the synthetic families with exactly known basin measure are independent of our method and can serve as a common test bed for other localization methods.

The portfolio of splitting driven by transient duration, active learning of basin boundaries, and a uniform share is never much worse than uniform sampling in the number of integrations, gains by factors of 3–7 on the rare hidden attractors of the benchmark that have a transient signal, changes the order of growth of the cost from p^{-1} to $p^{-0.45}$ where the signal is present on all scales, and gains by 10^3–10^4 on coupled copies of systems with hidden attractors. In wall time, the gain carries over where trajectories are bounded, as in the Chua circuits, and shrinks or vanishes where most trajectories escape within a few steps and a uniform point is cheap. The failures of the method are as informative as its gains: they identify the deceptive coordinates, the discrete escape times, the long boundaries, and the time-resolution limit that delineate its domain of applicability. The gains come from the transient duration as a reaction coordinate rather than from the particular way of climbing it: a single stagger-and-step chain on the same coordinate is as strong or stronger where the signal is present and fails where it is absent, so it is a natural arm for the portfolio, whose uniform share keeps the bound.

Beyond the method, the work leaves two tools and one object. The hiddenness measure h_ε grades hidden attractors by the scale at which the equilibria cease to lead to them; the verification protocol catches both transient chaos mistaken for an attractor and an attractor destroyed by a coarse integration step; and the enumeration of regimes with that protocol revealed a hidden chaotic attractor of two coupled Chua circuits that the single circuit does not have.

ACKNOWLEDGMENTS

This work received no external funding.

Use of generative AI. The AI assistant Claude (Anthropic) was used to help edit the text of the article and to write the code for the computations. The formulation of the problem, the design of the method and of the benchmark, the choice of the experiments, the analysis of the results, and the conclusions are the author's. All literature references were checked against the Crossref and Semantic Scholar databases. The author takes full responsibility for the content of the article.

AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Author Contributions

Maksim Elisov: Conceptualization (lead); Methodology (lead); Software (lead); Validation (lead); Formal analysis (lead); Investigation (lead); Data curation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are openly available in Zenodo at https://doi.org/10.5281/zenodo.23348470 (Ref. 107) and in GitHub at https://github.com/Hardos220/hidden-attractor-search. The repository contains the code (Python 3.14 with numba; Julia 1.13 with Attractors.jl for the independent check), the configurations of all experiments, the results in JSON format, and the scripts that produce the figures. Every result reported in the text is accompanied in the repository by a script that reproduces it.

Appendix A: Equations and parameters of the benchmark systems

Chua circuit:[2,24]

$$\dot{x} = \alpha\big(y - x - f(x)\big), \quad \dot{y} = x - y + z,$$
$$\dot{z} = -(\beta y + \gamma z),$$
$$f(x) = m_1 x + \tfrac{1}{2}(m_0 - m_1)\big(|x+1| - |x-1|\big).$$

The parameters $(\alpha, \beta, \gamma, m_0, m_1)$ are

set I: (8.4562, 12.0732, 0.0052, −0.1768, −1.1468),
set II: (8.4, 12, −0.005, −1.2, −0.05).

For set II, the known regimes are the cycle LC passing through $(-7.287, -2.773, 36.02)$ and the chaotic attractors passing through $(1.528, 0.283, -1.819)$ and $(-1.304, 0.029, 2.387)$. Two coupled circuits (Sec. V D): $\dot{x}_u = \alpha(y_u - x_u - f(x_u)) + \epsilon(\bar{x} - x_u)$, $\bar{x} = (x_1 + x_2)/2$, with the other equations unchanged, in the region B^2.

Systems without equilibria. NE4: $\dot{x} = -0.1y + a$, $\dot{y} = x + z$, $\dot{z} = xz - 3y$, $a = 1$. Wang–Chen #1:[14] $\dot{x} = y$, $\dot{y} = z$, $\dot{z} = -y + 3y^2 - x^2 - xz + a$, $a = -0.05$. Wei:[13] $\dot{x} = -y$, $\dot{y} = cx + z$, $\dot{z} = ay^2 + xz - d$, $(a, c, d) = (2, 1, 0.35)$. Jafari *et al.* 2026:[54] $\dot{x} = y$, $\dot{y} = z$, $\dot{z} = a_1 z + a_2 xy + a_3 yz + a_4 zx + a_5$, $(a_1, a_2, a_4, a_5) = (6, -1, -1, -1)$; a cycle at $a_3 = -1$ and chaos at $a_3 = -3.548$, with the state $(6.7445, 0.4214, -1.7846)$ on the attractor.

A single stable equilibrium. Wang–Chen #2:[14] $\dot{x} = yz + a$, $\dot{y} = x^2 - y$, $\dot{z} = 1 - 4x$, $a = 0.006$. SE3:[15] $\dot{x} = y$, $\dot{y} = z$, $\dot{z} = -3.4x - y - 4z + y^2 + xy$. SE11:[15] $\dot{x} = y$, $\dot{y} = -x + yz$, $\dot{z} = -y - 12z + x^2 + 9xz - 1$.

A line of equilibria. LE1:[16] $\dot{x} = y$, $\dot{y} = -x + yz$, $\dot{z} = -x - axy - bxz$, $(a, b) = (15, 1)$; the equilibria $(0, 0, z)$ are stable for $z < 0$.

Rabinovich system:[10] $\dot{x} = -\sigma(x - y) - ayz$, $\dot{y} = rx - y - xz$, $\dot{z} = -bz + xy$, $\sigma = -ar$; $(r, a, b) = (100, -0.009965, 0.077454)$ (hidden attractor) and $(6.8, -0.5, 0.99)$ (transient set[56]).

Negative control. The Lorenz system $\dot{x} = a(y - x)$, $\dot{y} = rx - y - xz$, $\dot{z} = xy - bz$ at $(a, r, b) = (4, 29, 2)$.

Maps. Huynh *et al.*:[20] $x' = x + y$, $y' = y - a|y| - xy + bx^2 - cy^2 + d$, $(a, b, c, d) = (0.01, 0.1, 2, 0.1)$. General family of Jiang *et al.*:[18] $x' = y$, $y' = a_1 x + a_2 y + a_3 x^2 + a_4 y^2 + a_5 xy + a_6$; the cases used and their coefficients $(a_1, \ldots, a_6)$ are listed below.

- NF I e: $(1, 0, 0, -0.3, 0.3, 2.98)$, box $[-8, 8]^2$.
- NF I f: $(1, 0, 0, 0.38, -0.38, -1.6)$, box $[-5, 5]^2$.
- NF I g: $(1, 0, 0, 0.12, -0.12, -3.62)$, box $[-8, 8]^2$.
- NF I i: $(1, 0, 0, -0.3, 0.3, 1.09)$, box $[-5, 5]^2$.
- SF I a: $(-0.33, 0.17, -0.48, 0.47, 0.01, -0.9)$, box $[-5, 5]^2$.
- SF I b: $(-0.84, 0, 0, 0.15, -0.15, -5.85)$, box $[-12, 12]^2$.
- SF I c: $(-0.99, 0, 0, 0.23, -0.23, -4.24)$, box $[-10, 10]^2$.
- SF I d: $(-1.29, 2, -0.35, -0.85, 1.2, 0.17)$, box $[-5, 5]^2$.

The same family in the form of Ref. 85: $x' = y$, $y' = x + a_1 x^2 + a_2 y^2 - a_3 xy - a_4$, with the parameter sets $(0.2, 0.71, 0.91, 1.14)$, $(0.51, 1, 1.51, 0.74)$, and $(0.6, 1, 1.6, 0.72)$ ("Jiang–Ouannas 1–3" in Table VIII).

Four-dimensional systems. Li and Sprott:[86] $\dot{x} = y - x$, $\dot{y} = -xz + u$, $\dot{z} = xy - a$, $\dot{u} = -by$ at $(a, b) = (6, 0.1)$, $(7, 0.1)$, and $(2.6, 0.44)$. Jahanshahi *et al.*:[87] $\dot{x} = y$, $\dot{y} = z$, $\dot{z} = w$, $\dot{w} = -aw + bx^2 - cy^2 + exy + fxz + g$, $(a, b, c, e, f, g) = (1.05, 0.7, 0.19, 1.37, 1.79, 1.15)$.

Synthetic families D1 and D2: Eqs. (4) and (6) with the base SE3, $B = [-10, 10]^3 \times [-1, 1]^m$, $m = 1$–11, integrated with the settings of SE3; in D2, the transient segment is extended by $\ln(\sqrt{m}/\rho)/\kappa + 5$ time units so that every escaping trajectory leaves before the observation window.

The integration parameters and the regions of initial conditions are listed in Table VIII; the control systems are included. Figure 9 shows the Chua circuit with parameter set I, Fig. 10 the chaos window of the Jafari 2026 system, Fig. 11 the level-B maps, and Fig. 12 the level-C system of Li and Sprott.

Appendix B: Verification episodes

This appendix documents the three cases mentioned in Sec. III C and the measured hiddenness.

Numerical "transient chaos" in the Chua circuit. With RK4 at $\Delta t = 0.01$, all 256 copies of a state on the hidden chaotic set of Chua I converge within $t \le 2 \times 10^4$ to a spurious periodic orbit of the same size (median time about 1500; Fig. 13a). At $\Delta t = 0.0025$, none of 96 copies leaves before $t = 10^4$, and DOP853 preserves the chaos up to $t = 4000$. At $\Delta t = 0.01$, the fractions of the hidden attractors are underestimated almost twofold (Fig. 13b). For a piecewise-linear system, this is an integrator artifact and the mirror image of the spurious "hidden Lorenz attractor":[55,56] there, transient chaos was mistaken for an attractor; here, an attractor is mistaken for transient chaos. The step-convergence check is therefore mandatory.

Basin fractions of the Chua circuit. Two independent implementations, Python (RK4, $\Delta t = 0.0025$) and Attractors.jl (Vern9), give identical fractions in the balls $r = 3$ and $r = 5$ and in Zelinka's box. Neither of them reproduces Table 1 of the review in Ref. 24 (Table IX). In the ball $r = 3$, the escape fraction is 0.19 in our computations versus 0.06 in the review, the fraction of F_0 is 0.44 versus 0.75, and that of each hidden attractor is 0.18 versus 0.095. The last pair is suggestive: our fractions of the hidden attractors at the coarse step $\Delta t = 0.01$ are also about 0.09 (Fig. 13b), so the review's values are consistent with the artifact above, whereas its escape and F_0 fractions remain unexplained. The basin structure in the plane $z = 0$ coincides with the review and with the catalog of Ref. 108 (Fig. 9).

Intermittency and transient sets. In NE4, the "second attractor" of a preliminary version of the protocol turned out to be intermittency of the single chaotic attractor; this was fixed by the window $T_{\mathrm{obs}} = 500 \gg 1/\lambda$. The transient chaos of the Lorenz system $(4, 29, 2)$ lives about 10^4 time units: all eight copies decay by $t \le 7 \times 10^4$.

TABLE VIII. Integration parameters and regions of initial conditions B. For flows, the lengths of the transient segment and of the observation window are given in time units, and for maps in iterations; R_{esc} is the escape threshold.

System	n	Integration	$T_{\mathrm{tr}}/T_{\mathrm{obs}}$	Region B	R_{esc}
Chua I	3	RK4, $\Delta t = 0.0025$	1000 / 100	$[-10,10]^2 \times [-15,15]$	100
Chua II	3	RK4, $\Delta t = 0.0025$	1000 / 100	$[-10,10]^2 \times [-15,15]$	100
NE4	3	RK4, $\Delta t = 0.001$	500 / 500	$[-40,20] \times [-30,50] \times [-80,220]$	2000
Wang–Chen #1	3	RK4, $\Delta t = 0.01$	300 / 100	$[-10,10]^3$	100
Wei (NE2)	3	RK4, $\Delta t = 0.01$	300 / 100	$[-10,10]^3$	100
Jafari 2026, $a_3 = -1$	3	RK4, $\Delta t = 0.01$	3000 / 200	$[-10,10]^3$	60
Jafari 2026, $a_3 = -3.548$	3	RK4, $\Delta t = 0.005$	1000 / 200	$[-10,10]^3$	60
Wang–Chen #2	3	RK4, $\Delta t = 0.01$	300 / 100	$[-10,10]^3$	100
SE3	3	RK4, $\Delta t = 0.01$	300 / 100	$[-10,10]^3$	100
SE11	3	RK4, $\Delta t = 0.01$	300 / 100	$[-10,10]^3$	100
LE1	3	RK4, $\Delta t = 0.01$	300 / 100	$[-10,10]^3$	100
Rabinovich, $r = 100$	3	RK4, $\Delta t = 0.002$	200 / 100	$[-60,60]^2 \times [-150,150]$	1000
Rabinovich, $r = 6.8$	3	RK4, $\Delta t = 0.005$	300 / 100	$[-10,10]^2 \times [-10,15]$	200
Lorenz $(4, 29, 2)$	3	RK4, $\Delta t = 0.005$	300 / 100	$[-20,20]^2 \times [0,50]$	500
D1, D2 (base SE3)	$3+m$	RK4, $\Delta t = 0.01$	≥ 300 / 100	$[-10,10]^3 \times [-1,1]^m$	100
Huynh 2019	2	iterations	20 000 / 5000	$[-3,3]^2$	100
Jiang NF I e	2	iterations	20 000 / 5000	$[-8,8]^2$	1000
Jiang NF I f	2	iterations	20 000 / 5000	$[-5,5]^2$	1000
Jiang NF I g	2	iterations	20 000 / 5000	$[-8,8]^2$	1000
Jiang NF I i	2	iterations	20 000 / 5000	$[-5,5]^2$	1000
Jiang SF I a	2	iterations	20 000 / 5000	$[-5,5]^2$	1000
Jiang SF I b	2	iterations	20 000 / 5000	$[-12,12]^2$	1000
Jiang SF I c	2	iterations	20 000 / 5000	$[-10,10]^2$	1000
Jiang SF I d	2	iterations	20 000 / 5000	$[-5,5]^2$	1000
Jiang–Ouannas 1–3	2	iterations	20 000 / 5000	$[-2,2]^2$	100
Hénon (control)	2	iterations	5000 / 2000	$[-2,2]^2$	50
Li–Sprott 4D, $a = 6$	4	RK4, $\Delta t = 0.005$	500 / 200	$[-15,15]^4$	500
Li–Sprott 4D, $a = 7$	4	RK4, $\Delta t = 0.005$	500 / 200	$[-15,15]^4$	500
Li–Sprott 4D, hyperchaos	4	RK4, $\Delta t = 0.005$	500 / 200	$[-15,15]^4$	500
Jahanshahi 4D	4	RK4, $\Delta t = 0.005$	500 / 200	$[-10,10]^4$	500
Coupled Chua II, $\epsilon = 0.1$	6	RK4, $\Delta t = 0.0025$	1000 / 100	$\left([-10,10]^2 \times [-15,15]\right)^2$	141

A protocol with an observation window shorter than the lifetime would accept this set as an attractor (Fig. 14). The Rabinovich set at $r = 6.8$ lives longer than 10^5 for six of eight copies, consistent with the estimate $\tau \sim 5 \times 10^7$ of Ref. 56; in the benchmark, it is used only as a control.

Hiddenness measure. Table X lists h_ε for the systems with equilibria. For SE3 and SE11, $h_\varepsilon = 0$ at all three scales. For Chua I, h_ε grows from 0 at $\varepsilon = 10^{-3}L$ through 0.003 ($10^{-2}L$) to 0.19 ($10^{-1}L$). For the Rabinovich system at $r = 100$, the values are 0, 0.03, and 0.65: the attractor is hidden only in the strict limit $\varepsilon \to 0$, and already at $\varepsilon = 10^{-2}L$, trajectories from the neighborhood of an equilibrium enter its basin. The binary definition thus does not distinguish "deeply" and "shallowly" hidden attractors, whereas h_ε does. For the Jafari chaos, whose copies with a perturbation of 10^{-3} escape (Fig. 14), all 32 copies with a perturbation of 10^{-6} stay on the attractor for 10^4 time units with RK4, and so do eight copies with DOP853, with finite-time exponents of 0.056 ± 0.011 and none below 0.01: it is an attractor with a basin thinner than a relative distance of 10^{-3} near the attractor, not a long transient.

A measured zero of h_ε is informative only relative to the number of starts n: with no hits, $h_\varepsilon < 3/n$ at the 95% level, so the zeros of Table X bound h_ε by the values given there, and for the rare targets with p comparable to $3/n$ the zero does not by itself separate "hidden" from "rare." For SE11, Chua II, and the rare family of the coupled Chua circuits, the measure was therefore remeasured with 2×10^5 starts per radius: no start reaches a hidden target at $10^{-3}L$ or $10^{-2}L$, so $h_\varepsilon < 1.5 \times 10^{-5}$ at the 95% level, below the basin measures of all three targets (Sec. V D).

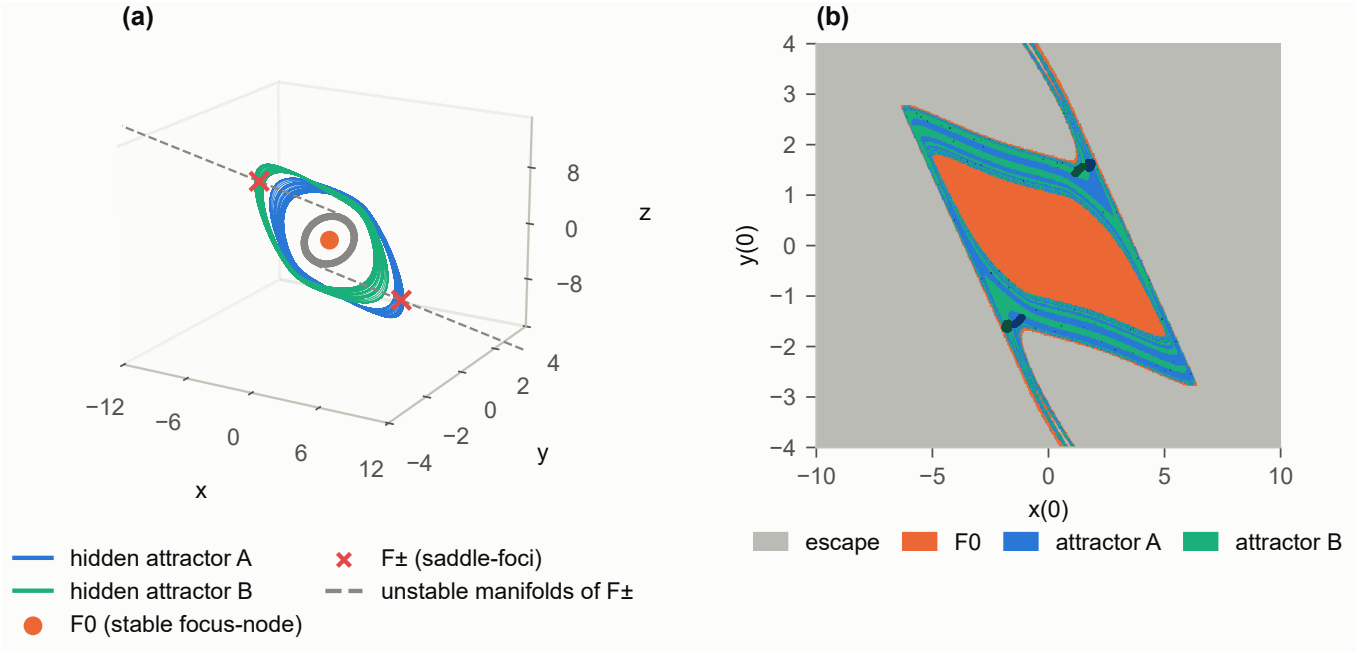


FIG. 9. Chua circuit, parameter set I. (a) Two symmetric hidden chaotic attractors A and B, the stable focus-node F_0 (dot), the saddle-foci $F_\pm$ (crosses), and their unstable manifolds. One branch of each manifold escapes to infinity and the other winds onto F_0, so the standard procedure does not find attractors A and B. (b) Basins in the plane $z(0) = 0$ (window $|y| \le 4$). Gray: escape to infinity; orange: F_0; blue and green: attractors A and B; dark points: the attractors in the layer $|z| < 0.15$.

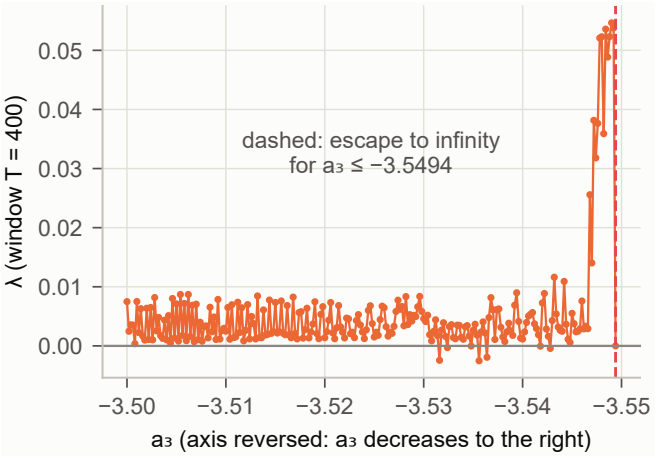


FIG. 10. System of Jafari *et al.*[54] without fixed and perpetual points: the Lyapunov exponent over windows $T = 400$ in the narrow chaos window near the boundedness limit (a_3 decreases to the right); for $a_3 \le -3.5494$, trajectories escape to infinity. The cycle at $a_3 = -1$ and the chaotic attractor at $a_3 = -3.548$ are shown in Fig. 1.

TABLE IX. Basin fractions of Chua I under uniform sampling in balls of radius $r = 3$ and $r = 5$ about the origin: escape to infinity (∞), the stable equilibrium F_0, and the two symmetric hidden attractors A and B. Sources: Table 1 of the review in Ref. 24 and two independent implementations, Python (RK4, $dt = 0.0025$, 10^5 points per ball) and Attractors.jl (Vern9, 10^4 points per ball). The two implementations agree with each other but not with the review. In the implementations, A and B are matched by the sign of the mean of x on the attractor; for the review, they are listed in the order of its Table 1. The fraction of unmatched trajectories in the Python run (0.06%) is omitted.

	$r = 3$				$r = 5$			
Source	∞	F_0	A	B	∞	F_0	A	B
Ref. 24, Table 1	0.06	0.75	0.09	0.10	0.23	0.50	0.14	0.13
Python, RK4	0.19	0.44	0.18	0.18	0.55	0.18	0.14	0.14
Attractors.jl	0.19	0.44	0.18	0.19	0.55	0.18	0.13	0.14

Appendix C: Classical heuristics

The classical heuristics behave on the benchmark in agreement with the review in Ref. 24 (Table XI). Perpetual points lead to the target in Wang–Chen #1, Wei, and Rabinovich. The Chua circuit has none: in the linear zones, the acceleration vanishes only at equilibria. They are also absent in SE3, SE11, and Wang–Chen #2, while in NE4 and LE1 they exist but do not lead to the target. Sampling on nullclines (coordinate-wise zero-velocity sets[38]) gives gains in hit frequency from 1.1 (SE11) and 2.6 (LE1) to 4.5 (Chua I). Connecting curves[35] gain a factor of 2 in Wang–Chen #1 and give no hits in SE3 and SE11. Continuation from the describing function finds the target in Chua I within 10 integrations but applies only to Lur'e-type systems.

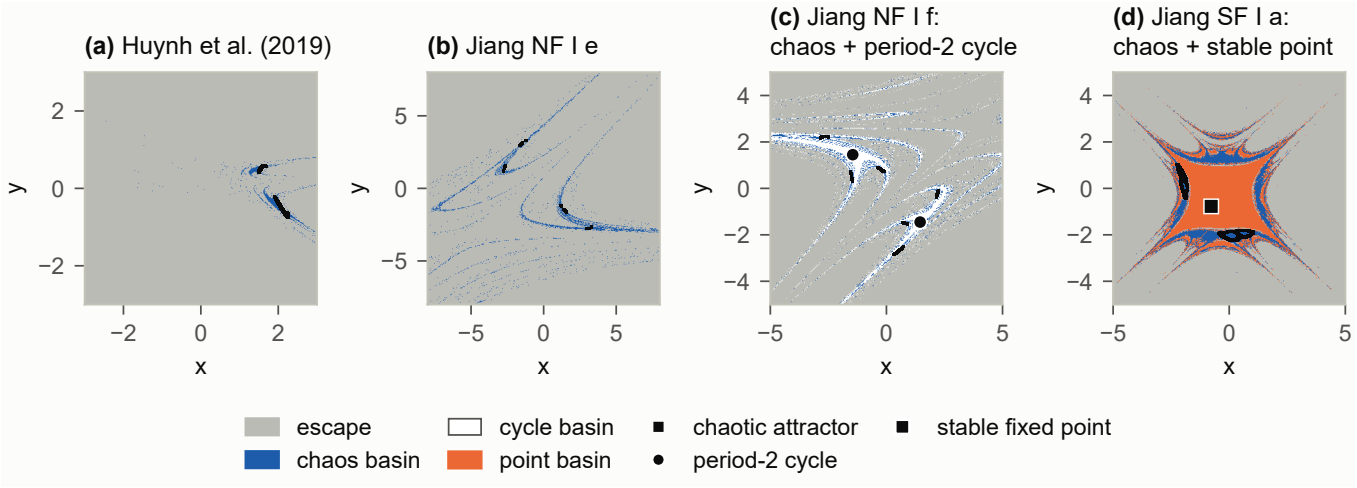


FIG. 11. Two-dimensional maps of level B: phase–basin portraits. (a) The map of Huynh *et al.*[20] without fixed points ($p = 0.0091$ in $[-3, 3]^2$). (b), (c) Cases NF I e and NF I f of the family of Jiang *et al.*[18] without fixed points; in NF I f, a chaotic attractor coexists with a period-2 cycle (markers). (d) Case SF I a with a single stable fixed point (square). Light gray: escape; dark blue: basin of the chaotic attractor; white: basin of the cycle; orange: basin of the fixed point; black: the attractor.

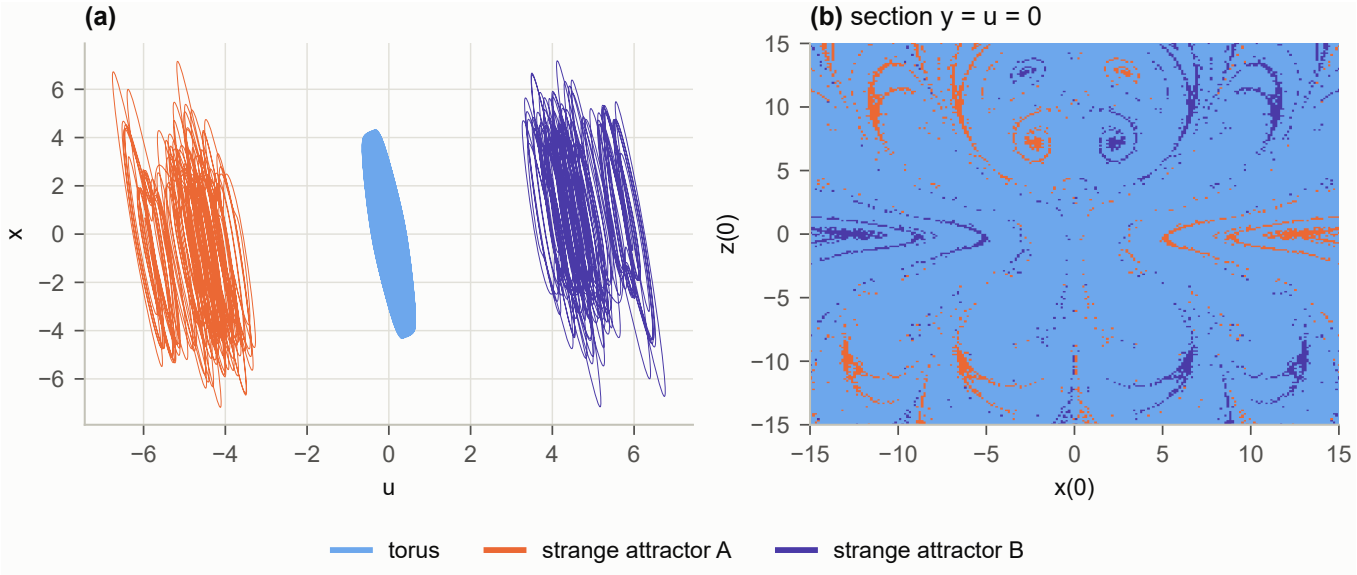


FIG. 12. Four-dimensional simplified Lorenz system[86] at $a = 6$, $b = 0.1$; there are no equilibria. (a) The torus (basin fraction 0.090) and a pair of symmetric strange attractors ($\lambda \approx 0.25$; 0.446 and 0.464) in the projection (u, x). (b) Basins in the section $y = u = 0$ of the box $[-15, 15]^4$; the torus occupies the "background," and the strange attractors form spiral tongues. Here the rare target is the torus, not the chaos.

TABLE X. Hiddenness measure h_ε: the fraction of trajectories started in the ε-neighborhoods of the equilibria (uniformly in the ball of radius ε about each equilibrium, plus the two points $e \pm \varepsilon u$ along every unstable eigendirection u) that enter the basin of the attractor, for $\varepsilon = 10^{-3}L$, $10^{-2}L$, and $10^{-1}L$, where L is the half-width of the region B; n is the number of starts per radius. An entry 0 means no hit among the n starts (95% upper bound $3/n$). For Chua I and Chua II, the two hidden attractors are listed separately; the values quoted in the text for Chua I are the sums over A and B. The Lorenz transient set is shown for comparison.

System, attractor	n	$\varepsilon = 10^{-3}L$	$10^{-2}L$	$10^{-1}L$
Chua I, A	1204	0	0.00249	0.0839
Chua I, B	1204	0	0.000831	0.101
Chua II, cycle LC	1202	0	0	0.403
Chua II, chaos	1202	0	0	0.0957
Wang–Chen #2	400	0	0	0.838
SE3	400	0	0	0
SE11	400	0	0	0
Rabinovich, $r = 100$	1202	0	0.0324	0.650
Lorenz $(4, 29, 2)$, transient set	1202	0	0	0.0408

TABLE XI. Classical heuristics versus uniform sampling on the level-A targets: the number of integrations to detection with probability 0.9, $N_{0.9} = \lceil \ln 0.1 / \ln(1-p) \rceil$, where p is the hit frequency of the sampling scheme. Uniform: sampling in B (p from 2×10^4 points; the geometric value, which differs slightly from the measured $N^0_{0.9}$ of Table II). CVS: sampling on the coordinate-wise zero-velocity sets (nullclines) $\dot{x}_i = 0$, up to 10^3 points on each, best coordinate. CC: sampling on connecting curves, 400 points. "Not found": no hit among the sampled points. PP: the number of perpetual points that lead to the target over the number of perpetual points found ("none": the system has no perpetual points).

Target	Uniform $N^0_{0.9}$	CVS $N_{0.9}$	CC $N_{0.9}$	PP hits/total
Rabinovich, $r = 100$	2	2	2	2/2
Wang–Chen #2	3	1	2	none
NE4	6	7	29	0/1
SE3	14	11	not found	none
Wang–Chen #1	22	19	11	1/2
Wei (NE2)	23	17	22	1/2
Chua I, chaos A[a]	136	14	172	none
Chua I, chaos B[a]	124	14	172	none
SE11	291	255	not found	none
LE1	1475	575	920	0/1

[a] The heuristics count hits on either of the two hidden attractors; the uniform cost for either attractor is 65.

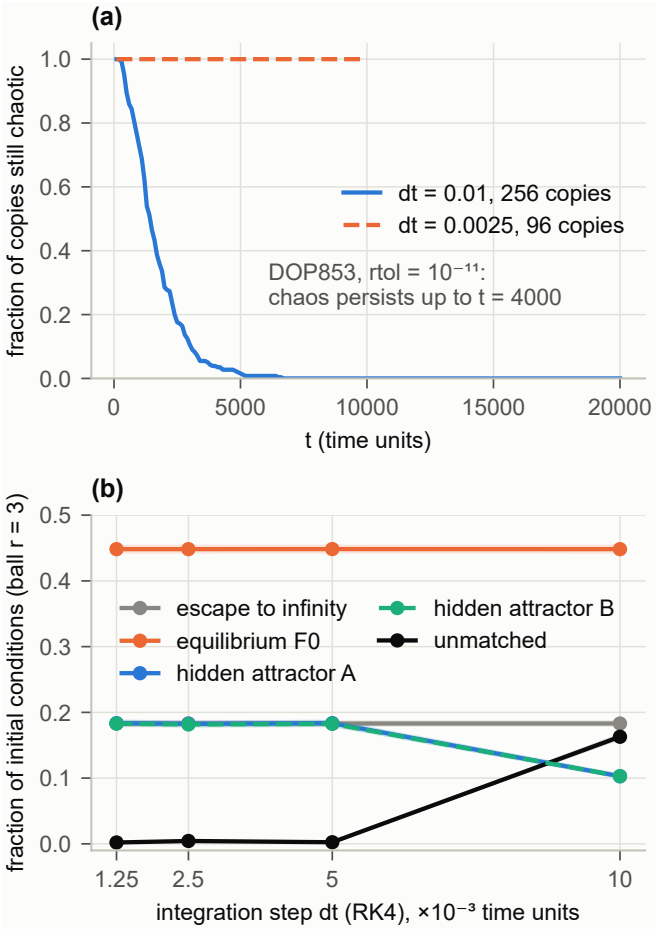


FIG. 13. Numerical "transient chaos" in Chua I. (a) The fraction of copies (initial perturbation 10^{-3}) remaining on the hidden chaotic set. With RK4 at $\Delta t = 0.01$, the chaos decays onto a spurious periodic orbit; at $\Delta t = 0.0025$, it persists to the end of the window. (b) Basin fractions versus the RK4 step (ball $r = 3$, 2×10^4 points, $T_{\mathrm{tr}} = 1000$). At $\Delta t = 0.01$, about 16% of trajectories land on the spurious orbit ("unmatched"), and the fractions of the hidden attractors are underestimated. For $\Delta t \le 0.005$, the fractions stabilize.

Appendix D: Ablation, baselines, and robustness

Ablation. The contributions of the components are shown in Fig. 15. On Chua II, the arms add up: the portfolio gives 7.4 times, versus 5.1 for splitting and 3.2 for active learning. On LE1, the active-learning arm hurts, and the gain of the portfolio is provided by the splitting arm. Without the uniform share, the portfolio is not faster (Chua II: 7.0; SE11: 3.1; LE1: 5.1) and loses where there is no signal (Jafari cycle: 0.27 against 0.43). On LE1, the lower interval bound for G without the uniform share is 1.59, whereas for the portfolio with $\theta = 0.2$, it is 3.76. Multidimensional "rarity" as a reaction coordinate produces heavy tails: the chains get stuck in atypical but nontarget regions, for example, in the basin of an equilibrium inside the D2 ball. The escape step as a coordinate

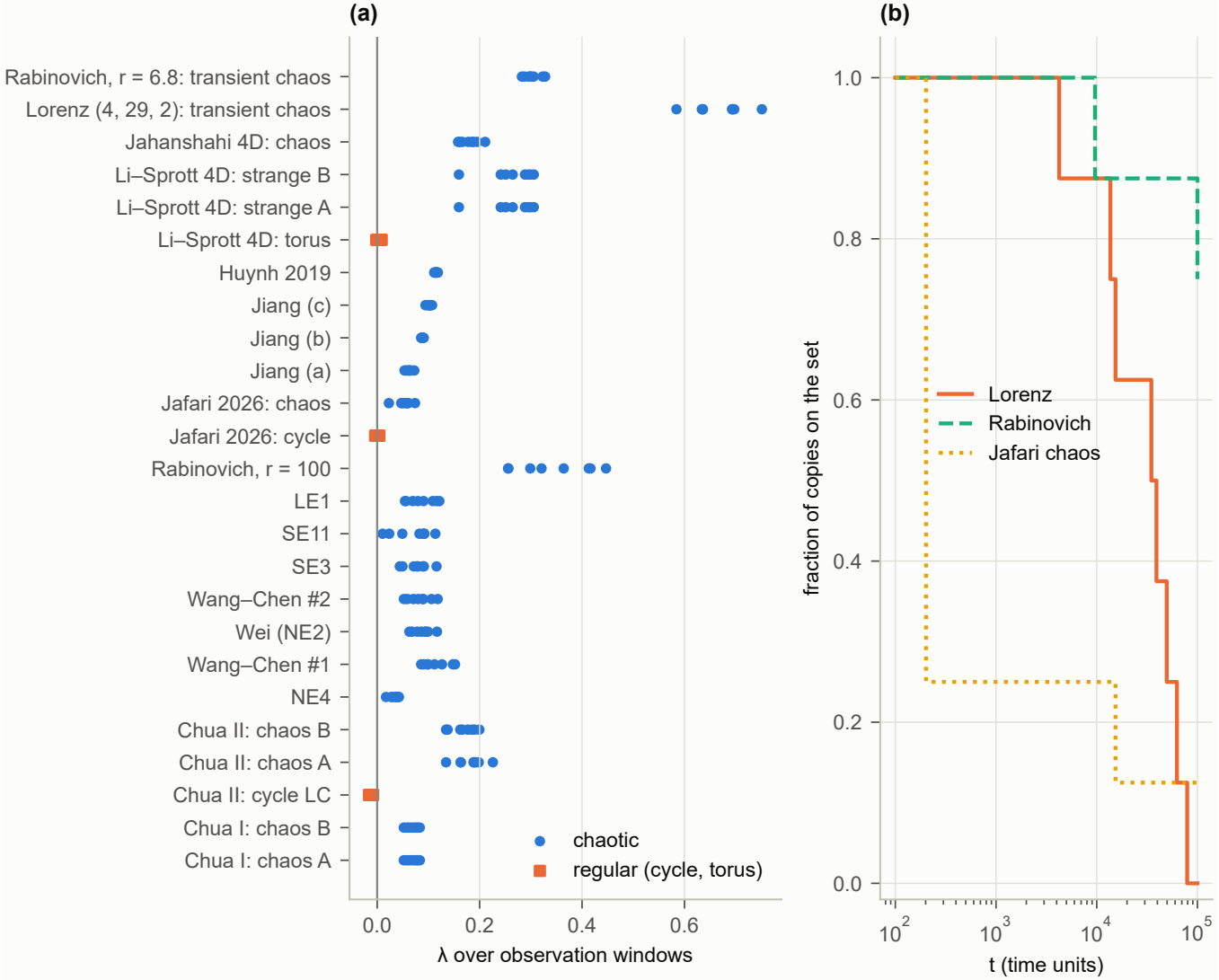


FIG. 14. Verification of the targets. (a) The largest Lyapunov exponent over consecutive observation windows for all targets of the benchmark; blue: chaotic regimes; orange: regular ones (cycle LC of Chua II, the Jafari cycle, the Li–Sprott torus). (b) Long lifetime check (eight copies with initial perturbation 10^{-3}). The Lorenz transient chaos decays completely. For the Rabinovich set at $r = 6.8$, six of eight copies survive to 10^5. For the fragile Jafari chaos, six of eight copies escape to infinity already by $t = 200$: with a relative perturbation of 10^{-3}, the copies leave its thin basin; with 10^{-6}, all of them stay on the attractor (Appendix B).

is equivalent to ξ everywhere except Chua II, where all trajectories are bounded and it gives 1.04 against 5.05. Two earlier strategies were also tested ($R = 20$): adaptive multilevel splitting driven by the novelty of the asymptotic fingerprint finds the target on Chua II in only 55% of runs, and MAP-Elites with the fingerprint as behavior descriptor has a median 5.6 times better than uniform sampling but a heavy tail, $G = 0.36$. Such tails are what the uniform share of the portfolio removes.

Differential-evolution baseline. The baseline is differential evolution over initial conditions with the cost function of Ref. 40, in the variant DE/rand/1/bin used there. The population has $N_P = 25$ members (half of the initial sample $n_0 = 50$ of the other strategies, but not fewer than 10), the differential weight is $F = 0.9$, and the crossover probability is $C_R = 0.3$, the values of Ref. 40. The initial population is drawn uniformly from the region B and consists of the first 25 points of the uniform stream of the run, so that it coincides with the first 25 points of uniform sampling (common random numbers). In every generation, for each member x_i, three other members x_{r_1}, x_{r_2}, x_{r_3}, distinct from one another and from x_i, are drawn without replacement; the mutant is $v = x_{r_3} + F\,(x_{r_1} - x_{r_2})$; the trial u_i takes each coordinate from v with probability C_R and from x_i otherwise, with one randomly chosen coordinate always taken from v; and u_i is clipped to B coordinate by coordinate. The 25 trials of a generation form one batch

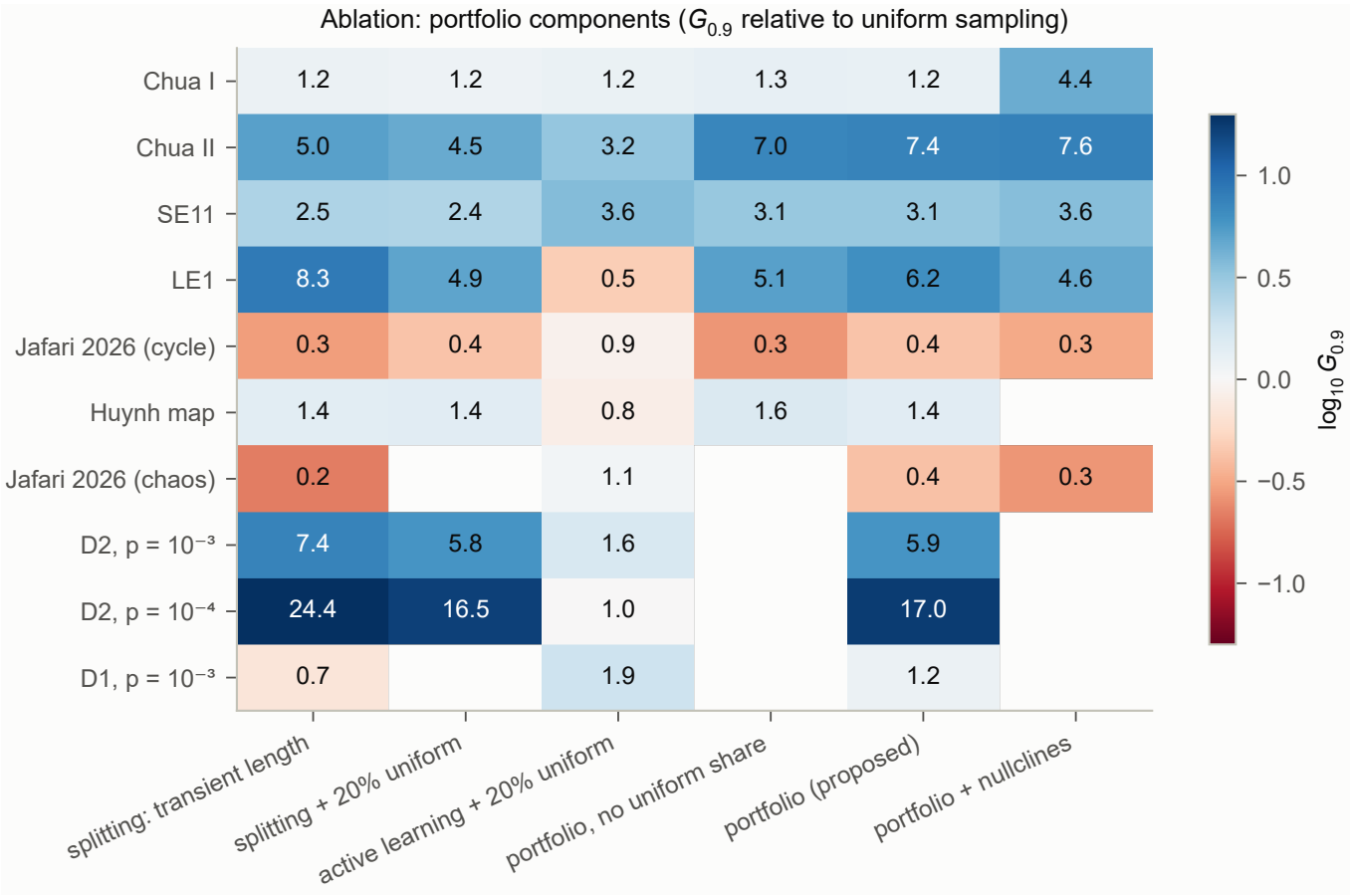


FIG. 15. Ablation: $G_{0.9}$ of the variants of the method relative to uniform sampling ($R = 100$; color: $\log_{10} G$; empty cells: the variant was not run). Rows: systems of levels A and B and synthetic families; columns: splitting alone and with a uniform share, active learning with a uniform share, the portfolio without a uniform share, the portfolio, and the portfolio with nullclines. Columns 1, 3, 5, and 6 repeat Table II; the new information is in columns 2 and 4 and in the D1 and D2 rows.

and are integrated with the full protocol ($T_{\mathrm{tr}} + T_{\mathrm{obs}}$ of Table VIII); u_i replaces x_i if its cost is not larger than that of x_i. The cost is computed from the trajectory fingerprint alone: with $r_{\max}$ the maximum of the Euclidean norm of the state over the observation window T_{obs} (after the transient T_{tr}), the cost is 0 if $d_{\mathrm{lo}} \le r_{\max} \le d_{\mathrm{hi}}$ and $r_{\max}$ otherwise, and it is 10^6 for a trajectory that escapes beyond R_{esc} or produces a non-finite state at any time. This is the function of Ref. 40, where the maximum is taken over the last 50 time units of a run of length 200 and the interval $[d_{\mathrm{lo}}, d_{\mathrm{hi}}] = [3, 11]$ was chosen empirically for the hidden attractor of the Chua circuit; the pair $(3, 11)$ remains the default of the implementation but is not used in the benchmark. Instead, the interval is set from the size of the target, which is the sense in which the baseline is "told the size": if r^* denotes the value of $r_{\max}$ in the fingerprint of the target reference (the mean over its 64 calibration copies; for a problem with several target regimes, the largest of their values), then $[d_{\mathrm{lo}}, d_{\mathrm{hi}}] = [0.5\, r^*, 1.5\, r^*]$. For Chua I, $r^* = 10.9$ and the interval is $[5.5, 16.4]$. The cost does not use the labels of the identification protocol, so the search does not distinguish the target from any other bounded regime whose size falls into the interval. Budget and detection are the same as for every other strategy: each integrated trial counts as one evaluation, the initial population included; a run has the per-problem budget of Sec. V A (5–15 times $N^0_{0.9}$ on levels A and B, from 10^3 integrations on Chua I to 4×10^4 on the Jafari chaos); and the first hit is the ordinal number, in the order of evaluation, of the first integrated point whose fingerprint the identification protocol matches to the target reference, whether or not that point is accepted into the population. The run stops at the end of the generation in which the hit occurs, and $N_{0.9}$ is the 0.9 quantile of the first-hit number over the R runs.

Bisection toward basin boundaries. A natural alternative that uses basin boundaries directly is bisection toward them, in the spirit of edge tracking[80,81] and of the straddle method.[83] A chain starts from a point with a long transient and the nearest point of another class and halves the segment between them 12 times; new chains start as old ones end. Bisection is a strong baseline on systems with smooth boundaries (Table XII): it gives 4.2

TABLE XII. Bisection toward basin boundaries ($R = 50$) and the portfolio ($R = 100$, from Tables II and III): gains $G_{0.9}$ with 95% bootstrap intervals. Bold: values whose lower interval bound exceeds 1. A lower bound of 0 means that some bootstrap resamples did not reach 90% success.

Problem	Bisection	Portfolio
Chua I	**1.30** [1.06, 1.95]	1.24 [0.94, 1.77]
Chua II	**3.14** [1.69, 5.12]	**7.36** [6.09, 10.2]
SE11	4.17 [0, 4.64]	**3.10** [2.46, 3.70]
LE1	**6.02** [4.47, 9.57]	**6.24** [3.76, 10.0]
Jafari 2026, cycle	0.96 [0.53, 1.61]	0.43 [0.29, 0.78]
Huynh 2019	1.05 [0.37, 2.21]	**1.40** [1.16, 2.54]
Jafari 2026, chaos	1.18 [0.77, 2.06]	0.42 [0.29, 0.58]
D2, $p = 10^{-3}$	1.64 [1.00, 2.52]	**5.88** [5.02, 8.10]
D2, $p = 10^{-4}$	1.29 [0, 1.82]	**17.0** [13.6, 22.2]
D1, $p = 10^{-3}$	**3.62** [2.84, 4.89]	1.19 [0.86, 1.58]

on SE11 (portfolio: 3.1), 6.0 on LE1 (6.2), 1.3 on Chua I (1.2), and 3.6 on D1 at $p = 10^{-3}$ (1.2). It loses where the signal is graded over many scales: 3.1 on Chua II (7.4) and 1.6 and 1.3 on D2 at $p = 10^{-3}$ and 10^{-4} (5.9 and 17). On the problems without a signal, bisection stays near 1 (0.96–1.18), whereas the portfolio pays for its arms (0.42–0.43). On the coupled copies of SE11, it does not scale (Sec. V D). Bisection as a fourth arm of the portfolio, with fixed or adaptive shares, helps only where bisection alone is strong (in $R = 50$ runs: D1 at $p = 10^{-3}$, 1.7 against 1.2; LE1, 8.6 against 6.2) and takes budget from splitting on the problems with the largest gains (three coupled copies of SE11 at $R = 20$: 1360 against 1570). We therefore retain the three-arm portfolio.

Stagger-and-step. The splitting arm is a multichain relative of the stagger-and-step method,[84] which locates chaotic saddles by climbing the escape time of a single trajectory with random perturbations of log-uniformly distributed size. We ran a stagger-and-step search with the same reaction coordinate: a single chain keeps the point with the largest ξ, proposes a perturbation in a random direction with a size log-uniform between 10^{-3} and 1 of the region width, accepts it when ξ increases, and restarts from a uniform point after 30 consecutive rejections; full integration and the budgets of Table II were used ($R = 50$; three coupled copies of SE11: $R = 20$). The results are in Table XIII. Where the coordinate carries a signal, stagger-and-step is the strongest strategy of the paper: 1.7 on Chua I, 10 on Chua II, and from 2.1 to 155 on D2 at $p = 10^{-2}$–10^{-5}, with a scaling exponent $s = 0.32$ [0.28, 0.37] against 0.45 for the portfolio and 0.41 for splitting; on SE11, LE1, and three coupled copies of SE11, it is comparable with the portfolio (2.9, 5.9, and 735 against 3.1, 6.2, and 1435). Where the coordinate is deceptive or uninformative, it fails, and without a uniform share it falls below the bound of the portfolio: 0.16 on the Jafari cycle, 0.26 on the Jafari chaos, and 0.93 on the Huynh map, and on D1 it finds the target within six times the uniform cost in only 60% of the runs.

TABLE XIII. Stagger-and-step[84] on the reaction coordinate ξ against splitting and the portfolio: gains $G_{0.9}$ relative to the uniform reference of Tables II–IV, with 95% bootstrap intervals over the stagger runs. Bold: lower interval bound above 1; "—" with a percentage: fewer than 90% of the runs found the target within the budget.

Problem	R	Stagger-and-step	Splitting	Portfolio
Chua I	50	**1.67** [1.02, 2.20]	1.20	1.24
Chua II	50	**10.1** [8.80, 13.1]	5.05	7.36
SE11	50	**2.90** [1.71, 3.69]	2.53	3.10
LE1	50	**5.85** [3.77, 8.86]	8.29	6.24
Jafari 2026, cycle	50	0.16 [0.14, 0.26]	0.29	0.43
Huynh 2019	50	0.93 [0.63, 1.45]	1.36	1.40
Jafari 2026, chaos	50	0.26 [0.23, 0.36]	0.22	0.42
D1, $p = 10^{-3}$	20	— (60%)	0.69	1.19
D2, $p = 10^{-2}$	50	**2.14** [1.93, 2.90]	2.29	2.27
D2, $p = 10^{-3}$	50	**6.99** [5.32, 9.07]	7.37	5.87
D2, $p = 10^{-4}$	50	**34.5** [31.3, 50.9]	24.4	17.0
D2, $p = 10^{-5}$	50	**155** [120, 187]	93.1	72.0
3 coupled SE11	20	**735** [237, 1357]	682	1435

The gain thus comes from the reaction coordinate rather than from the way of climbing it, and the contribution of the portfolio is the bound. A stagger-and-step chain is a natural arm of the portfolio that would combine its efficiency with the guarantee; we leave this combination to future work.

Robustness. The hyperparameters of the portfolio were not tuned to the problems: everywhere $\theta = 0.2$, $p_0 = 0.2$, $\sigma_0 = 0.1$, and $n_0 = 50$. Each of them was varied separately on three problems with a gain ($R = 50$; Table XIV). No variant brings the gain below 1: on Chua II, it lies in the range 5.2–7.5; on D2, 14.7–21.8; and on LE1, 3.1–9.2. A larger initial sample ($n_0 = 100$) worsens the gain on targets with $p \sim 10^{-3}$, where the initial sample is a noticeable part of the cost, and improves it on D2 ($p = 10^{-4}$). Scaling both integration windows by 0.5 and 2 on Chua II, SE11, LE1, and D2 ($p = 10^{-4}$) changed the cost of uniform sampling by at most 12% and kept the gains within a factor of 1.7 of the base values: 7.0 and 6.3 on Chua II (base 7.4), 3.8 and 3.6 on SE11 (3.1), 3.6 and 6.7 on LE1 (6.2), and 18 and 18 on D2 (17).

Appendix E: Real systems in enlarged regions

To test whether the gain grows on real systems as p decreases, we enlarged the region of initial conditions k times about its center, $B_k = kB$ with $k = 2$–16, keeping the escape radius at least twice the distance to the farthest corner. If the basin lay inside B, its measure in B_k would be $p_{k=1}/k^3$. It is not (Table XV, Fig. 16a): for SE11, p falls from 7.7×10^{-3} only to 2.4×10^{-4} at $k = 16$, against 1.9×10^{-6} expected, and the other systems behave similarly. The basins of these hidden attractors extend far beyond B, so their rarity is a property of the chosen

TABLE XIV. Robustness of the portfolio to hyperparameters: $G_{0.9}$ with 95% bootstrap intervals (variants: $R = 50$; base: $R = 100$). In each column, one parameter is changed relative to the base set $\theta = 0.2$, $p_0 = 0.2$, $\sigma_0 = 0.1$, $n_0 = 50$.

Problem	Base	$\theta = 0.1$	$\theta = 0.3$	$p_0 = 0.1$	$p_0 = 0.3$	$\sigma_0 = 0.3$	$n_0 = 100$
Chua II	7.36 [6.09, 10.2]	6.90 [5.27, 10.4]	5.70 [4.58, 8.24]	7.53 [6.31, 11.7]	6.32 [5.23, 8.87]	5.23 [4.23, 8.29]	5.23 [4.27, 7.79]
LE1	6.24 [3.76, 10.0]	6.86 [4.07, 13.3]	6.48 [3.34, 10.9]	9.16 [1.96, 14.7]	6.85 [4.96, 10.8]	3.15 [2.39, 6.85]	3.12 [2.11, 5.45]
D2, $n = 6$, $p = 10^{-4}$	17.0 [13.6, 22.2]	14.7 [10.6, 24.1]	18.5 [13.2, 24.7]	21.8 [14.4, 36.2]	18.8 [13.9, 27.8]	18.2 [13.7, 26.5]	20.7 [11.1, 28.2]

region rather than of the attractor. Continuation of the hidden attractors in a parameter gives the same conclusion: in Chua II, the Wang–Chen and Wei systems, the basin fraction stays at 10^{-3}–0.7 up to the crisis at which the attractor disappears, and in LE1 it falls to 2.4×10^{-4} at most. Over the enlarged regions, the gain grows as p decreases on three of the five systems (Fig. 16b): on SE11, from 3.1 to 24; on LE1, from 6.2 to 16; and on Chua I, from 1.2 to 6.3. The effective exponents of these growths, $\ln G / \ln(p_{k=1}/p)$, are 0.4–0.7, with the caveat that the enlargement changes the geometry of the basin as well as its measure. On Chua II, the gain falls from 7.4 to 0.71 at $k = 4$, and splitting alone falls to 0.14, because the longest transients belong to trajectories that return slowly from far away rather than to the neighborhood of the target. On the Jafari cycle, where there is no signal, the portfolio stays at 0.24–0.43, within the guarantee.

Appendix F: Discretization limit at very small p

Figure 17 shows the gain on D2 down to $p = 10^{-7}$ and the diagnostic variants. Two alternative explanations of the breakdown below $p = 10^{-5}$ were ruled out. Lowering the step floor of the splitting chains tenfold, or removing it, leaves every run unchanged: the floor is never reached. A second move that redraws a random subset of coordinates from the uniform distribution, akin to the componentwise Metropolis algorithm of the original subset simulation,[64] helps only partly: at $p = 10^{-6}$, splitting then succeeds in 50% of runs and the portfolio in 68%, but at 10^{-7} there is still no gain. The same move doubles the gain at $p = 10^{-4}$ (53 for splitting and 35 for the portfolio); it is a natural extension of the method but is not used elsewhere in this paper. Reducing the integration step restores the order-of-growth gain, as stated in Sec. V C; these gains are counted in integrations, each of which is 4 or 16 times longer, so the gain in computing time is correspondingly smaller; the test demonstrates the mechanism, not a practical speedup. The level threshold also explains why one informative point is not enough: the level ξ_l is the tenth largest value in the reference sample, and when fewer than ten points are informative, the threshold is $\xi_l = 0$ (all escapes at the first step tie), the selection is empty, and the chains restart; the gain at $p = 10^{-5}$ is carried by the runs whose reference sample contains several informative points. Figure 18 shows the dependence on dimension at fixed $p = 10^{-4}$.

Appendix G: Channel diagnostics

The prediction of the cost model was tested on a uniform sample of 2×10^4 points per system (Fig. 19). Channel B works when nontarget points with long transients converge spatially to the target. For Chua II, the median distance $d(q)$ decreases with the upper-set fraction q as $0.43 \to 0.19 \to 0.12$; for LE1, as $0.53 \to 0.22 \to 0.13$; for SE11, as $0.47 \to 0.19 \to 0.15$; and for D2, as $0.74 \to 0.60 \to 0.53$ (at $q = 0.2$, 0.04, and 0.008); the splitting gain here is 2.5–8.3 times. For the Jafari cycle ($0.99 \to 1.35$) and for D1 ($1.26 \to 1.38 \to 1.41$), $d(q)$ grows, and there is no gain. At the same time, in D1, the fraction of the target in the upper sets grows: the regions of long trajectories are scattered, and there is no path to the target through them. The AUC of the coordinate ξ alone (0.88–0.97 for all systems except the Jafari cycle, 0.61) does not separate these cases, whereas $d(q)$ does. For the Huynh map, there is a gradient over only one decade (d drops to 0.22 and grows again); at $p \approx 0.009$, it barely pays back the initial sample and the levels.

Appendix H: Implementation details of the two-stage configuration

The own computations of the arms (nearest-neighbor queries of the active-learning arm, the chains, and the matching) take 1–8% of the wall time on the Chua systems with up to two copies, 28% on three uncoupled copies of Chua II, and 20–60% on the cheap problems, where an integration costs microseconds. Brute-force neighbor search for $n \geq 6$ reduces the queries by a factor of 1.6–2.6 relative to a tree (G_T from 16 to 29 on three coupled copies of SE11). The cap on the training sample of the classifier (the last 2×10^4 evaluated points) removes the growth of the overhead in long runs: on three uncoupled copies of Chua II, it lowers the overhead from 51% to 28% and, by training the classifier on recent points only, also lowers the median cost from 1.4×10^5 to 8.6×10^4 integrations and raises the number of runs that succeed within 4×10^5 integrations from 8 to 9 of 10. A wider batch, with two or four times more chains and boundary points to occupy more cores, does not pay: the number of evaluations grows by a factor of 1.5–3, whereas the time per evaluation decreases by about 10% only. The gains in time of Table VI were measured with four threads per process and those of Table VII on one thread; with the

TABLE XV. Real systems in the enlarged regions kB ($R = 50$; $k = 1$: $R = 100$): basin measure, cost of uniform sampling, and gains $G_{0.9}$ with 95% bootstrap intervals. Bold: values whose lower interval bound exceeds 1; a lower bound of 0 means that some bootstrap resamples did not reach 90% success; "—" with a percentage: fewer than 90% of the runs found the target.

System	k	p	$N^0_{0.9}$	Splitting	Portfolio
SE11	1	0.0077	317	**2.53** [1.95, 2.86]	**3.10** [2.46, 3.70]
SE11	2	0.0023	811	**3.91** [3.03, 4.66]	**4.93** [3.96, 6.01]
SE11	4	6.1×10^{-4}	3858	**11.8** [6.38, 19.4]	**17.5** [8.95, 26.8]
SE11	8	4.5×10^{-4}	5191	**9.92** [5.54, 14.5]	**9.19** [5.87, 17.0]
SE11	16	2.4×10^{-4}	9351	**15.8** [9.96, 21.8]	**24.2** [13.8, 34.5]
LE1	1	0.0013	1617	**8.29** [6.35, 12.3]	**6.24** [3.76, 10.0]
LE1	2	5.3×10^{-4}	4540	**8.81** [6.08, 17.2]	**7.51** [2.41, 14.4]
LE1	4	3.4×10^{-4}	5911	**11.1** [7.35, 23.5]	**7.19** [2.99, 10.5]
LE1	8	1.8×10^{-4}	13 229	**18.3** [12.9, 27.9]	**15.9** [10.8, 23.0]
Chua I	1	0.033	66	1.20 [0.92, 1.77]	1.24 [0.94, 1.77]
Chua I	2	0.0083	267	**2.30** [1.61, 3.23]	**2.61** [1.87, 3.84]
Chua I	4	0.0010	2230	**6.48** [3.32, 10.7]	**5.58** [3.80, 9.62]
Chua I	8	1.8×10^{-4}	11 169	**6.16** [4.16, 12.5]	**6.30** [4.04, 13.9]
Chua II	1	0.0026	818	**5.05** [4.13, 7.07]	**7.36** [6.09, 10.2]
Chua II	2	7.7×10^{-4}	2891	**2.61** [1.71, 4.11]	**3.74** [2.67, 7.55]
Chua II	4	2.1×10^{-4}	8988	0.14 [0.10, 0.25]	0.71 [0.57, 1.36]
Jafari cycle	1	0.010	219	0.29 [0.19, 0.53]	0.43 [0.29, 0.78]
Jafari cycle	2	0.0016	1245	0.22 [0, 0.30]	0.34 [0, 0.57]
Jafari cycle	4	2.2×10^{-4}	10 061	— (88%)	0.24 [0, 0.35]

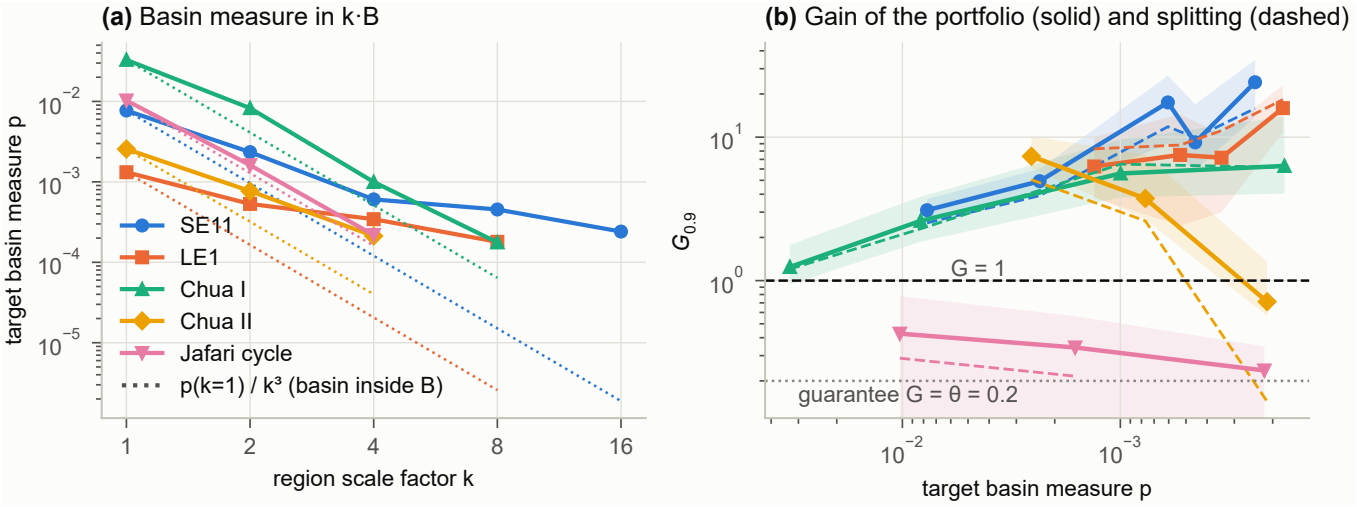


FIG. 16. Real systems in enlarged regions kB. (a) Basin measure of the hidden target versus k; dotted lines: $p_{k=1}/k^3$, the measure if the basin lay inside B. (b) Gain of the portfolio (solid, with 95% intervals) and of splitting (dashed) versus the basin measure, which decreases to the right. Dashed horizontal line: $G = 1$; dotted: the guarantee $G = \theta = 0.2$.

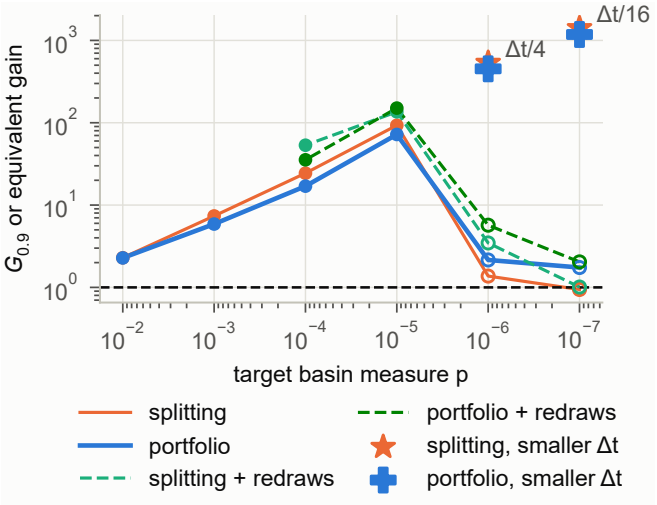


FIG. 17. Family D2 ($n = 6$): gain down to $p = 10^{-7}$. Filled markers: $G_{0.9}$; open markers: equivalent gain $-\ln(1-f)/(pN_{\max})$ from the fraction f of runs that found the target within the budget $N_{\max}$. Dashed lines: the variant with an additional move that redraws a random subset of coordinates (the variants with a lowered step floor coincide with the base curves and are not shown). Stars and crosses: splitting and the portfolio with the integration step reduced 4 times ($p = 10^{-6}$) and 16 times ($p = 10^{-7}$), with $N^0_{0.9} = \ln 10/p$.

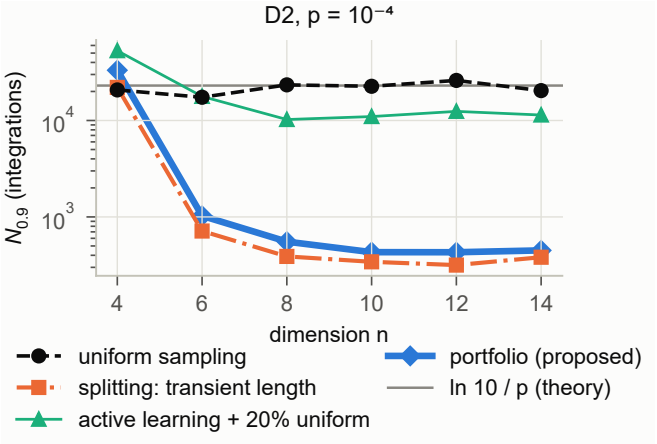


FIG. 18. Detection cost versus the phase-space dimension (family D2, $p = 10^{-4}$, $R = 100$). Thin gray line: $\ln 10/p$. At $n = 4$, the reaction coordinate is uninformative because of the discretization limit; for $n \geq 6$, the portfolio gain is 17–61 times.

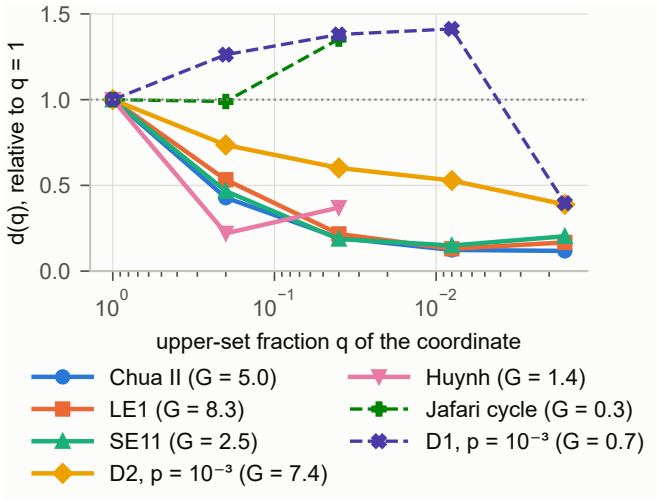


FIG. 19. Diagnostics of channel B from a uniform sample: the median distance from nontarget points of the upper set of the coordinate ξ to the target (normalized to the value at $q = 1$) versus the fraction q, which decreases to the right. The legend gives the splitting gain $G_{0.9}$. A decreasing $d(q)$ (solid: Chua II, LE1, SE11, D2) comes with a splitting gain (Tables II and III), and an increasing one (dashed: Jafari cycle, D1) with its absence; for the Huynh map, $d(q)$ decreases over one decade only, and the gain is small.

batch of 500 evaluations used here, the portfolio parallelizes better than uniform sampling (its time to detection falls 4.0-fold from one to four threads on the coupled Chua circuits, against 1.6-fold for the cost per uniform evaluation), so the four-thread gains are larger, and the one-thread values are the conservative ones.

[1]G. A. Leonov and N. V. Kuznetsov, "Hidden attractors in dynamical systems. From hidden oscillations in Hilbert–Kolmogorov, Aizerman, and Kalman problems to hidden chaotic attractor in Chua circuits," Int. J. Bifurcation Chaos **23**, 1330002 (2013).
[2]G. Leonov, N. Kuznetsov, and V. Vagaitsev, "Localization of hidden Chua's attractors," Phys. Lett. A **375**, 2230–2233 (2011).
[3]N. V. Kuznetsov, "Theory of hidden oscillations and stability of control systems," J. Comput. Syst. Sci. Int. **59**, 647–668 (2020).
[4]G. Leonov, N. Kuznetsov, and V. Vagaitsev, "Hidden attractor in smooth Chua systems," Physica D **241**, 1482–1486 (2012).
[5]N. V. Stankevich, N. V. Kuznetsov, G. A. Leonov, and L. O. Chua, "Scenario of the Birth of Hidden Attractors in the Chua Circuit," Int. J. Bifurcation Chaos **27**, 1730038 (2017).
[6]N. Kuznetsov, T. Mokaev, V. Ponomarenko, E. Seleznev, N. Stankevich, and L. Chua, "Hidden attractors in Chua circuit: mathematical theory meets physical experiments," Nonlinear Dyn. **111**, 5859–5887 (2023).
[7]B. Andrievsky, E. Kudryashova, N. Kuznetsov, O. Kuznetsova, and G. Leonov, "Hidden attractors in aircraft control systems with saturated inputs," (2017), arXiv:1711.10302.
[8]V. O. Bragin, V. I. Vagaitsev, N. V. Kuznetsov, and G. A. Leonov, "Algorithms for finding hidden oscillations in nonlinear systems. The Aizerman and Kalman conjectures and Chua's circuits," J. Comput. Syst. Sci. Int. **50**, 511–543 (2011).
[9]G. Chen, N. V. Kuznetsov, G. A. Leonov, and T. N. Mokaev, "Hidden Attractors on One Path: Glukhovsky–Dolzhansky, Lorenz, and Rabinovich Systems," Int. J. Bifurcation Chaos **27**, 1750115 (2017).
[10]N. V. Kuznetsov, G. A. Leonov, T. N. Mokaev, A. Prasad, and M. D. Shrimali, "Finite-time Lyapunov dimension and hidden attractor of the Rabinovich system," Nonlinear Dyn. **92**, 267–285 (2018).
[11]N. V. Kuznetsov, T. N. Mokaev, O. A. Kuznetsova, and E. V. Kudryashova, "The Lorenz system: hidden boundary of practical stability and the Lyapunov dimension," Nonlinear Dyn. **102**, 713–732 (2020).
[12]S. Jafari, J. Sprott, and S. M. R. Hashemi Golpayegani, "Elementary quadratic chaotic flows with no equilibria," Phys. Lett. A **377**, 699–702 (2013).
[13]Z. Wei, "Dynamical behaviors of a chaotic system with no equilibria," Phys. Lett. A **376**, 102–108 (2011).
[14]X. Wang and G. Chen, "A chaotic system with only one stable equilibrium," Commun. Nonlinear Sci. Numer. Simul. **17**, 1264–1272 (2012).
[15]M. Molaie, S. Jafari, J. C. Sprott, and S. M. R. H. Golpayegani, "Simple chaotic flows with one stable equilibrium," Int. J. Bifurcation Chaos **23**, 1350188 (2013).
[16]S. Jafari and J. Sprott, "Simple chaotic flows with a line equilibrium," Chaos Solitons Fractals **57**, 79–84 (2013).
[17]S. Jafari, J. C. Sprott, V. T. Pham, C. Volos, and C. Li, "Simple chaotic 3D flows with surfaces of equilibria," Nonlinear Dyn. **86**, 1349–1358 (2016).
[18]H. Jiang, Y. Liu, Z. Wei, and L. Zhang, "Hidden chaotic attractors in a class of two-dimensional maps," Nonlinear Dyn. **85**, 2719–2727 (2016).
[19]C. Wang and Q. Ding, "A New Two-Dimensional Map with Hidden Attractors," Entropy **20**, 322 (2018).
[20]V. V. Huynh, A. Ouannas, X. Wang, V. T. Pham, X. Q. Nguyen, and F. E. Alsaadi, "Chaotic Map with No Fixed Points: Entropy, Implementation and Control," Entropy **21**, 279 (2019).
[21]M.-F. Danca and N. Kuznetsov, "Hidden chaotic sets in a Hopfield neural system," Chaos Solitons Fractals **103**, 144–150 (2017).
[22]D. Dudkowski, S. Jafari, T. Kapitaniak, N. Kuznetsov, G. Leonov, and A. Prasad, "Hidden attractors in dynamical systems," Phys. Rep. **637**, 1–50 (2016).
[23]S. Jafari, J. C. Sprott, and F. Nazarimehr, "Recent new examples of hidden attractors," Eur. Phys. J. Spec. Top. **224**, 1469–1476 (2015).
[24]X. Guan and Y. Xie, "A review on methods for localization of hidden attractors," Nonlinear Dyn. **113**, 22223–22255 (2025).
[25]U. Feudel, "Complex dynamics in multistable systems," Int. J. Bifurcation Chaos **18**, 1607–1626 (2008).
[26]A. N. Pisarchik and U. Feudel, "Control of multistability," Phys. Rep. **540**, 167–218 (2014).
[27]A. Chudzik, P. Perlikowski, A. Stefanski, and T. Kapitaniak, "Multistability and rare attractors in van der Pol–Duffing oscillator," Int. J. Bifurcation Chaos **21**, 1907–1912 (2011).
[28]X. Yue, G. Lv, and Y. Zhang, "Rare and hidden attractors in a periodically forced Duffing system with absolute nonlinearity," Chaos Solitons Fractals **150**, 111108 (2021).
[29]N. Kuznetsov, G. Leonov, and V. Vagaitsev, "Analytical-numerical method for attractor localization of generalized Chua's system," IFAC Proc. Vol. **43**, 29–33 (2010).
[30]N. Kuznetsov, O. Kuznetsova, G. Leonov, T. Mokaev, and N. Stankevich, "Hidden attractors localization in Chua circuit via the describing function method," IFAC-PapersOnLine **50**, 2651–2656 (2017).
[31]A. Prasad, "Existence of Perpetual Points in Nonlinear Dynamical Systems and Its Applications," Int. J. Bifurcation Chaos **25**, 1530005 (2015).
[32]D. Dudkowski, A. Prasad, and T. Kapitaniak, "Perpetual points and hidden attractors in dynamical systems," Phys. Lett. A **379**, 2591–2596 (2015).
[33]D. Dudkowski, A. Prasad, and T. Kapitaniak, "Perpetual Points: New Tool for Localization of Coexisting Attractors in Dynamical Systems," Int. J. Bifurcation Chaos **27**, 1750063 (2017).
[34]R. Gilmore, J. M. Ginoux, T. Jones, C. Letellier, and U. S. Freitas, "Connecting curves for dynamical systems," J. Phys. A: Math. Theor. **43**, 255101 (2010).
[35]X. Guan and Y. Xie, "Connecting curve: A new tool for locating hidden attractors," Chaos **31**, 113143 (2021).
[36]N. Naseri, S. Ambigapathy, M. S. Kafraj, F. Ghassemi, K. Rajagopal, and S. Jafari, "Connecting Curves as a Tool to Localize Hidden Attractors in a New Chaotic Hyperjerk System with No Equilibria," Int. J. Bifurcation Chaos **31**, 2150230 (2021).
[37]W. Song, Y. Zeng, Z. Li, and M. Wang, "'Connecting surfaces' can locate hidden attractors," Eur. Phys. J. Plus **138**, 168 (2023).
[38]P. Godara, D. Dudkowski, A. Prasad, and T. Kapitaniak, "New topological tool for multistable dynamical systems," Chaos **28**, 111101 (2018).
[39]M. F. Danca, N. V. Kuznetsov, and G. Chen, "Approximating hidden chaotic attractors via parameter switching," Chaos **28**, 013127 (2018).
[40]I. Zelinka, "Evolutionary identification of hidden chaotic attractors," Eng. Appl. Artif. Intell. **50**, 159–167 (2016).
[41]I. Zelinka, "Unconventional Algorithms and Hidden Chaotic Attractors," in *Chaotic Systems with Multistability and Hidden Attractors*, Emergence, Complexity and Computation (Springer, Cham, 2021) pp. 429–457.
[42]P. J. Menck, J. Heitzig, N. Marwan, and J. Kurths, "How basin stability complements the linear-stability paradigm," Nat. Phys. **9**, 89–92 (2013).
[43]G. Datseris and A. Wagemakers, "Effortless estimation of basins of attraction," Chaos **32**, 023104 (2022).
[44]G. Datseris, K. Luiz Rossi, and A. Wagemakers, "Framework for global stability analysis of dynamical systems," Chaos **33**, 073151 (2023).
[45]M. Stender and N. Hoffmann, "bSTAB: an open-source software for computing the basin stability of multi-stable dynamical systems," Nonlinear Dyn. **107**, 1451–1468 (2022).
[46]M. Gelbrecht, J. Kurths, and F. Hellmann, "Monte Carlo basin bifurcation analysis," New J. Phys. **22**, 033032 (2020).
[47]C. S. Hsu, *Cell-to-cell mapping: A method of global analysis for nonlinear systems* (Springer-Verlag, New York, 1987).
[48]M. Dellnitz and A. Hohmann, "A subdivision algorithm for the computation of unstable manifolds and global attractors," Nu-

mer. Math. **75**, 293–317 (1997).
[49]P. Brzeski, J. Wojewoda, T. Kapitaniak, J. Kurths, and P. Perlikowski, "Sample-based approach can outperform the classical dynamical analysis - experimental confirmation of the basin stability method," Sci. Rep. **7**, 6121 (2017).
[50]G. Datseris, A. Morr, M. Fadera, and J. Kurths, "Global continuation as a complement to traditional continuation and bifurcation analysis," (2026), arXiv:2607.09332.
[51]P. Haerter, A. Wagemakers, A. Daza, and M. A. F. Sanjuán, "Bayesian Basin Tracking: Efficient Global Continuation of Multistable Dynamical Systems," (2026), arXiv:2607.14762.
[52]F. Nazarimehr, B. Saedi, S. Jafari, and J. C. Sprott, "Are Perpetual Points Sufficient for Locating Hidden Attractors?" Int. J. Bifurcation Chaos **27**, 1750037 (2017).
[53]S. Jafari, F. Nazarimehr, J. C. Sprott, and S. M. R. H. Golpayegani, "Limitation of Perpetual Points for Confirming Conservation in Dynamical Systems," Int. J. Bifurcation Chaos **25**, 1550182 (2015).
[54]S. Jafari, A. Soheylifar, A. Bayani, K. Rajagopal, and J. C. Sprott, "Hidden Attractor in a Dynamical System Without Fixed Points and Perpetual Points," Int. J. Bifurcation Chaos **36**, 2630021 (2026).
[55]B. Munmuangsaen and B. Srisuchinwong, "A hidden chaotic attractor in the classical Lorenz system," Chaos Solitons Fractals **107**, 61–66 (2018).
[56]J. Sprott and B. Munmuangsaen, "Comment on 'A hidden chaotic attractor in the classical Lorenz system'," Chaos Solitons Fractals **113**, 261–262 (2018).
[57]M. F. Danca, "Hidden transient chaotic attractors of Rabinovich–Fabrikant system," Nonlinear Dyn. **86**, 1263–1270 (2016).
[58]J. A. Yorke and E. D. Yorke, "Metastable chaos: The transition to sustained chaotic behavior in the Lorenz model," J. Stat. Phys. **21**, 263–277 (1979).
[59]C. Grebogi, E. Ott, and J. A. Yorke, "Crises, sudden changes in chaotic attractors, and transient chaos," Physica D **7**, 181–200 (1983).
[60]Y.-C. Lai and T. Tél, *Transient Chaos: Complex Dynamics on Finite Time Scales*, Applied Mathematical Sciences, Vol. 173 (Springer, New York, 2011).
[61]P. Glasserman, P. Heidelberger, P. Shahabuddin, and T. Zajic, "Multilevel Splitting for Estimating Rare Event Probabilities," Oper. Res. **47**, 585–600 (1999).
[62]F. Cérou and A. Guyader, "Adaptive Multilevel Splitting for Rare Event Analysis," Stoch. Anal. Appl. **25**, 417–443 (2007).
[63]F. Cérou, A. Guyader, and M. Rousset, "Adaptive multilevel splitting: Historical perspective and recent results," Chaos **29**, 043108 (2019).
[64]S. K. Au and J. L. Beck, "Estimation of small failure probabilities in high dimensions by subset simulation," Probab. Eng. Mech. **16**, 263–277 (2001).
[65]R. Rubinstein, "The Cross-Entropy Method for Combinatorial and Continuous Optimization," Methodol. Comput. Appl. Probab. **1**, 127–190 (1999).
[66]J. Bect, D. Ginsbourger, L. Li, V. Picheny, and E. Vazquez, "Sequential design of computer experiments for the estimation of a probability of failure," Stat. Comput. **22**, 773–793 (2012).
[67]B. Echard, N. Gayton, and M. Lemaire, "AK-MCS: An active learning reliability method combining Kriging and Monte Carlo Simulation," Struct. Saf. **33**, 145–154 (2011).
[68]J. Lehman and K. O. Stanley, "Abandoning Objectives: Evolution Through the Search for Novelty Alone," Evol. Comput. **19**, 189–223 (2011).
[69]J.-B. Mouret and J. Clune, "Illuminating search spaces by mapping elites," (2015), arXiv:1504.04909.
[70]J. K. Pugh, L. B. Soros, and K. O. Stanley, "Quality Diversity: A New Frontier for Evolutionary Computation," Front. Robot. AI **3** (2016), 10.3389/frobt.2016.00040.
[71]A. Cully and Y. Demiris, "Quality and Diversity Optimization: A Unifying Modular Framework," IEEE Trans. Evol. Comput. **22**, 245–259 (2018).
[72]F. Ragone, J. Wouters, and F. Bouchet, "Computation of extreme heat waves in climate models using a large deviation algorithm," Proc. Natl. Acad. Sci. U.S.A. **115**, 24–29 (2018).
[73]T. Lestang, F. Ragone, C.-E. Bréhier, C. Herbert, and F. Bouchet, "Computing return times or return periods with rare event algorithms," J. Stat. Mech.: Theory Exp. **2018**, 043213 (2018).
[74]S. Baars, D. Castellana, F. W. Wubs, and H. A. Dijkstra, "Application of adaptive multilevel splitting to high-dimensional dynamical systems," J. Comput. Phys. **424**, 109876 (2021).
[75]X. S. Wang, S. A. Moore, J. D. Turner, and B. P. Mann, "A model-free sampling method for basins of attraction using hybrid active learning (HAL)," Commun. Nonlinear Sci. Numer. Simul. **112**, 106551 (2022).
[76]X. W. Cao, X. L. Ru, and G. Yan, "Active learning attraction basins of dynamical system," Chin. Phys. B **34**, 058901 (2025).
[77]Y. Du, Q. Li, H. Fan, M. Zhan, J. Xiao, and X. Wang, "Inferring attracting basins of power system with machine learning," Phys. Rev. Res. **6**, 013181 (2024).
[78]J. Shena, K. Kaloudis, C. Merkatas, and M. A. F. Sanjuán, "On the approximation of basins of attraction using deep neural networks," (2021), arXiv:2109.06564.
[79]M. Gameiro, B. Gelb, W. Kalies, M. Kramár, K. Mischaikow, and P. Tatasciore, "Data-Driven Identification of Attractors Using Machine Learning," SIAM J. Appl. Dyn. Syst. **25**, 836–860 (2025).
[80]J. D. Skufca, J. A. Yorke, and B. Eckhardt, "Edge of Chaos in a Parallel Shear Flow," Phys. Rev. Lett. **96**, 174101 (2006).
[81]T. M. Schneider, B. Eckhardt, and J. A. Yorke, "Turbulence Transition and the Edge of Chaos in Pipe Flow," Phys. Rev. Lett. **99**, 034502 (2007).
[82]J. Vollmer, T. M. Schneider, and B. Eckhardt, "Basin boundary, edge of chaos and edge state in a two-dimensional model," New J. Phys. **11**, 013040 (2009).
[83]H. E. Nusse and J. A. Yorke, "A procedure for finding numerical trajectories on chaotic saddles," Physica D **36**, 137–156 (1989).
[84]D. Sweet, H. E. Nusse, and J. A. Yorke, "Stagger-and-step method: Detecting and computing chaotic saddles in higher dimensions," Phys. Rev. Lett. **86**, 2261–2264 (2001).
[85]A. Ouannas, X. Wang, A. A. Khennaoui, S. Bendoukha, V. T. Pham, and F. E. Alsaadi, "Fractional Form of a Chaotic Map without Fixed Points: Chaos, Entropy and Control," Entropy **20**, 720 (2018).
[86]C. Li and J. C. Sprott, "Coexisting Hidden Attractors in a 4-D Simplified Lorenz System," Int. J. Bifurcation Chaos **24**, 1450034 (2014).
[87]H. Jahanshahi, M. Shahriari-Kahkeshi, R. Alcaraz, X. Wang, V. Singh, and V. T. Pham, "Entropy Analysis and Neural Network-Based Adaptive Control of a Non-Equilibrium Four-Dimensional Chaotic System with Hidden Attractors," Entropy **21**, 156 (2019).
[88]G. Datseris, "DynamicalSystems.jl: A Julia software library for chaos and nonlinear dynamics," J. Open Source Softw. **3**, 598 (2018).
[89]V. Lucarini and T. Bódai, "Edge states in the climate system: exploring global instabilities and critical transitions," Nonlinearity **30**, R32–R66 (2017).
[90]G. Benettin, L. Galgani, A. Giorgilli, and J. M. Strelcyn, "Lyapunov Characteristic Exponents for smooth dynamical systems and for hamiltonian systems; a method for computing all of them. Part 1: Theory," Meccanica **15**, 9–20 (1980).
[91]S. K. Lam, A. Pitrou, and S. Seibert, "Numba: A LLVM-based Python JIT compiler," in *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC* (2015) pp. 1–6.
[92]E. Hairer, S. P. Nørsett, and G. Wanner, *Solving Ordinary Differential Equations I: Nonstiff Problems* (Springer, Berlin, 1993).
[93]D. Lucente, C. Herbert, and F. Bouchet, "Committor Functions for Climate Phenomena at the Predictability Margin: The Ex-

ample of El Niño–Southern Oscillation in the Jin and Timmermann Model," J. Atmos. Sci. **79**, 2387–2400 (2022).
[94]I. Papaioannou, W. Betz, K. Zwirglmaier, and D. Straub, "MCMC algorithms for Subset Simulation," Probab. Eng. Mech. **41**, 89–103 (2015).
[95]B. Settles, *Active Learning* (Springer, Cham, 2012).
[96]T. Hesterberg, "Weighted Average Importance Sampling and Defensive Mixture Distributions," Technometrics **37**, 185–194 (1995).
[97]B. A. Huberman, R. M. Lukose, and T. Hogg, "An Economics Approach to Hard Computational Problems," Science **275**, 51–54 (1997).
[98]C. P. Gomes and B. Selman, "Algorithm portfolios," Artif. Intell. **126**, 43–62 (2001).
[99]B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap* (Chapman and Hall/CRC, New York, 1994).
[100]K. Murali and M. Lakshmanan, "Bifurcation and chaos of the sinusoidally-driven Chua's circuit," Int. J. Bifurcation Chaos **01**, 369–384 (1991).
[101]J. Pathak, Z. Lu, B. R. Hunt, M. Girvan, and E. Ott, "Using machine learning to replicate chaotic attractors and calculate Lyapunov exponents from data," Chaos **27**, 121102 (2017).
[102]Z. Lu, B. R. Hunt, and E. Ott, "Attractor reconstruction by machine learning," Chaos **28**, 061104 (2018).
[103]A. Haluszczynski and C. Räth, "Good and bad predictions: Assessing and improving the replication of chaotic attractors by means of reservoir computing," Chaos **29**, 103143 (2019).
[104]J. Herteux and C. Räth, "Breaking symmetries of the reservoir equations in echo state networks," Chaos **30**, 123142 (2020).
[105]A. Flynn, V. A. Tsachouridis, and A. Amann, "Multifunctionality in a reservoir computer," Chaos **31**, 013125 (2021).
[106]T. Kabayama, M. Komuro, K. Aihara, and K. Nakajima, "Pseudo-Lorenz attractors in echo state networks," Jpn. J. Ind. Appl. Math. **42**, 1459–1473 (2025).
[107]M. Elisov, "hidden-attractor-search: code, experiments, and results for 'Searching for hidden attractors as a rare-event problem: An adaptive strategy portfolio with a worst-case guarantee'," Zenodo (2026).
[108]A. Wagemakers, "Basins of Attraction: A Dynamical Zoo," Int. J. Bifurcation Chaos **35**, 2530024 (2025).